

0_10xMouse_RA_SoloTE

August 31, 2026

```
[1]: library(Seurat)
library(ggplot2)
library(ggpubr)
library(rcartocolor)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
library(clustree)
getwd()

dataset_id <- "10xMouse_RA"
dir.create("figures_10xMouse_RA")
dir.create("data_10xMouse_RA")
```

Loading required package: SeuratObject

Loading required package: sp

Attaching package: 'SeuratObject'

The following objects are masked from 'package:base':

intersect, t

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

Loading required package: ggraph

Attaching package: 'ggraph'

The following object is masked from 'package:sp':

```
geometry
```

```
'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'
```

```
Warning message in dir.create("figures_10xMouse_RA"):
```

```
"'figures_10xMouse_RA' already exists"
```

```
Warning message in dir.create("data_10xMouse_RA"):
```

```
"'data_10xMouse_RA' already exists"
```

```
[2]: colorDict = c("Differentiating"="#88535A",  
                  "2CLC"="#EF8264",  
                  "Pluripotent"="#F2CC8F")  
  
getwd()
```

```
'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'
```

1 Pre

```
[ ]: conversionTable <- read.table("annotation/annotation_mm10_conversion_withAge.  
  ↪tsv") # created with annotation_scripts/create_annotations_human.Rmd  
head(conversionTable)
```

```
[5]: SoloTE_path <- "/mnt/volume_1p5T/results/SoloTEout/10xMouse_RA/LIF/  
  ↪LIF_SoloTE_output/LIF_legacytes_MATRIX"  
  
# get filtered barcodes  
STAR_path <- "/mnt/TEresults/snakefile_results/results/STARoutdir/10xMouse_RA/  
  ↪LIF/best_Solo.out/Gene"  
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1_  
  ↪# read barcodes selected by STARsolo  
  
# 2CLC sample  
legacyTEmatrix <- Seurat::ReadMtx(mtx = paste0(SoloTE_path, "/matrix.mtx"),  
  cells = paste0(SoloTE_path, "/barcodes.tsv"),
```

```

features = paste0(SoloTE_path, "/features.tsv"))
# read matrix

legacyTEmatrix <- legacyTEmatrix[,filteredBarcodes]

# select TEs
TEs <- grep("SoloTE", rownames(legacyTEmatrix), value = T)
locusTEs <- grep("chr", TEs, value = T)
# subset the matrix keeping only TEs
TEmatrix <- legacyTEmatrix[locusTEs,]
# remove "SoloTE" from the name of the TEs

rownames(TEmatrix) <- gsub("SoloTE\\|", "", rownames(TEmatrix))

# rownames(TEmatrix) <- gsub("\\|", "-", rownames(TEmatrix))
# rownames(TEmatrix) <- gsub("\\_", "-", rownames(TEmatrix))
# rownames(TEmatrix) <- gsub("\\|?", "", rownames(TEmatrix))

# table(rownames(TEmatrix) %in% conversionTable$soloteID)
# # transform into Stellarscope IDs
# rownames(TEmatrix) <-
#   conversionTable$stellarscopeID[match(rownames(TEmatrix),
#   conversionTable$soloteID)]

nCells <- ncol(TEmatrix)
thrMinCells <- round(nCells * 0.02)
# create Seurat object with shallow filtering of TEs expressed in at least 50
#   cells and cells expressing at least 50 TEs
objTE <- Seurat::CreateSeuratObject(TEmatrix, project = "SoloTE",
  min.cells = thrMinCells, min.features = 50)
objTE
thrMinCells

```

Warning message:

"Feature names cannot have underscores ('_'), replacing with dashes ('-')"

Warning message:

"Feature names cannot have pipe characters ('|'), replacing with dashes ('-')"

An object of class Seurat

47725 features across 1542 samples within 1 assay

Active assay: RNA (47725 features, 0 variable features)

1 layer present: counts

31

```

[6]: objTE_SoloTE <- objTE#[,filteredBarcodes] # keep only filtered barcodes
objTE_SoloTE

```

An object of class Seurat

47725 features across 1542 samples within 1 assay
Active assay: RNA (47725 features, 0 variable features)
1 layer present: counts

```
[7]: # select genes
genes <- setdiff(rownames(legacyTEmatrix), TEs)
# subset the matrix keeping only genes
Genematrix <- legacyTEmatrix[genes,]
# create Seurat object with shallow filtering of genes expressed in at least 50
  ↪ cells and cells expressing at least 100 genes
Gene <- Seurat::CreateSeuratObject(Genematrix, project = "2CLC",
                                   min.cells = thrMinCells, min.features = 100)

objGenes_SoloTE <- Gene #[,filteredBarcodes] # keep only filtered barcodes
objGenes_SoloTE
```

An object of class Seurat
14316 features across 1542 samples within 1 assay
Active assay: RNA (14316 features, 0 variable features)
1 layer present: counts

```
[8]: # QC TEs

options(repr.plot.width=7, repr.plot.height=6)

objTE_SoloTE@meta.data$nCount_TE <- objTE_SoloTE@meta.data$nCount_RNA
objTE_SoloTE@meta.data$nFeature_TE <- objTE_SoloTE@meta.data$nFeature_RNA

# Visualize QC metrics as a violin plot
VlnPlot(objTE_SoloTE, features = c("nCount_RNA", "nFeature_RNA"), ncol = 2,
        cols = colorDict, pt.size = 0, alpha = 0.5) + geom_hline(yintercept =
  ↪ 400)
summary(objTE_SoloTE$nCount_RNA)
summary(objTE_SoloTE$nFeature_RNA)
```

Warning message:

"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."

Warning message:

"The `slot` argument of `FetchData()` is deprecated as of SeuratObject
5.0.0.

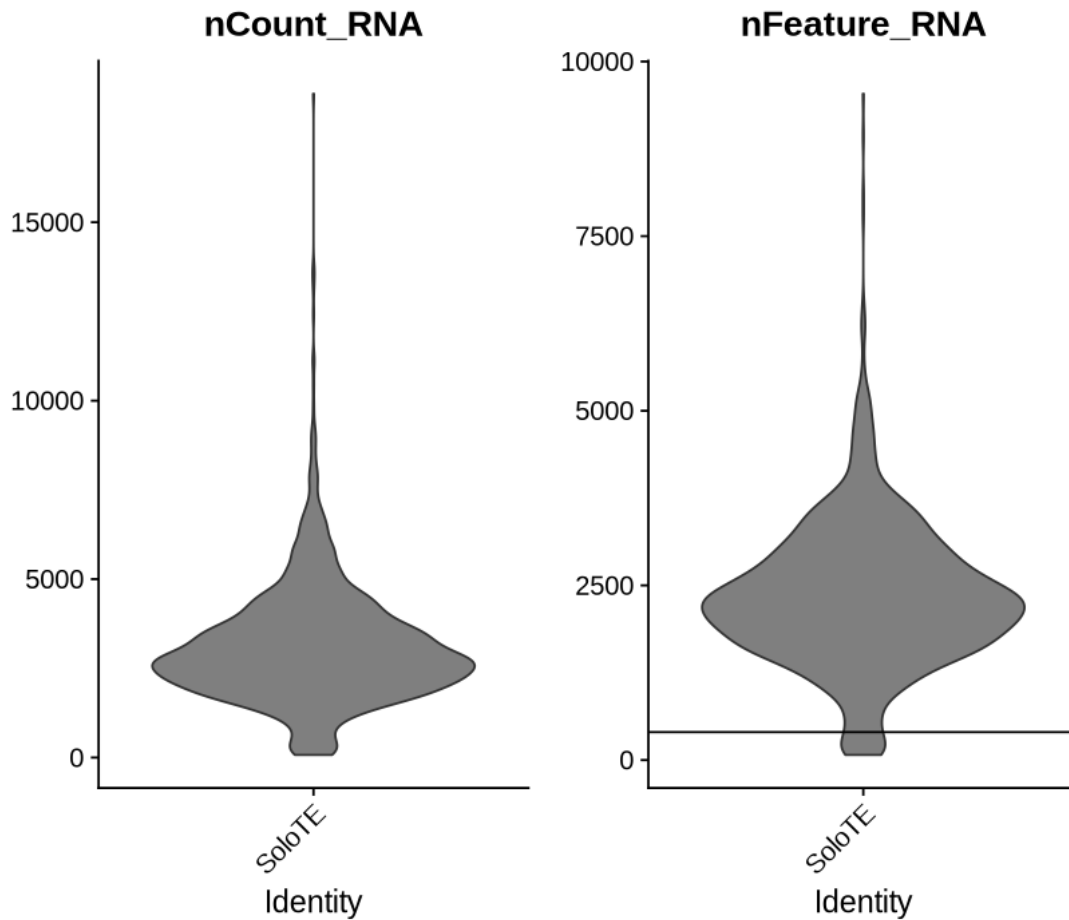
Please use the `layer` argument instead.

The deprecated feature was likely used in the [Seurat](#)
package.

Please report the issue at
<https://github.com/satijalab/seurat/issues>."

```
Warning message:
"`PackageCheck()` was deprecated in SeuratObject 5.0.0.
Please use `rlang::check_installed()` instead.
The deprecated feature was likely used in the Seurat
package.
Please report the issue at
<https://github.com/satijalab/seurat/issues>."
Warning message:
"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
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data's fill values."
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the
data's fill values."
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
74	2101	2802	3099	3777	18599
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
73	1712	2260	2337	2876	9538



```
[9]: # QC genes
options(repr.plot.width=8, repr.plot.height=5)

VlnPlot(objGenes_SoloTE, features = c( "nCount_RNA", "nFeature_RNA" ), ncol = 2,
      cols = alpha(colorDict, 0.5), pt.size = 0.05, alpha = 0.5) +
  geom_hline(yintercept = 2000)

summary(objGenes_SoloTE$nFeature_RNA)
summary(objGenes_SoloTE$nCount_RNA)
```

Warning message:

"Default search for "data" layer in "RNA" assay yielded no results; utilizing "counts" layer instead."

Warning message:

"No shared levels found between `names(values)` of the manual scale and the

data's `fill` values."

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data's `fill` values."

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data's `fill` values."

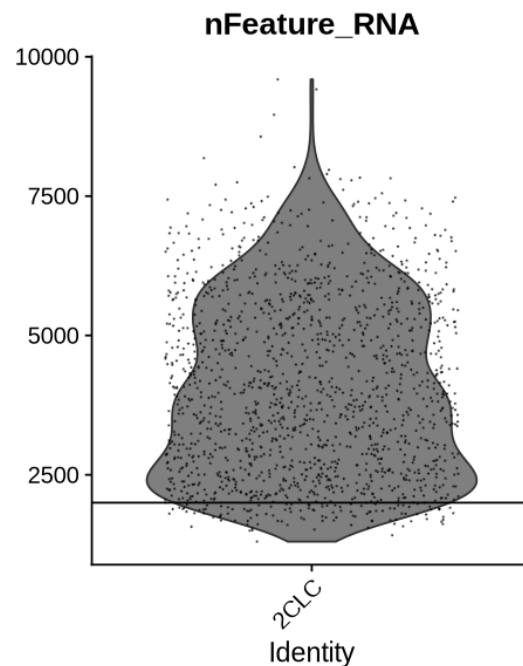
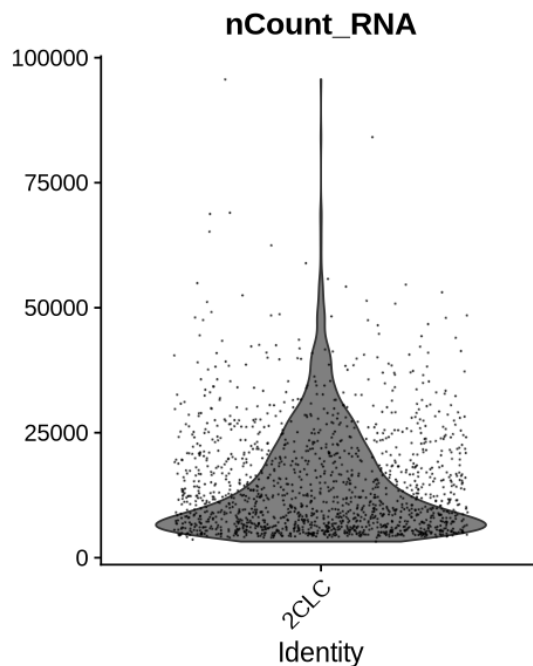
Warning message:

"No shared levels found between ``names(values)`` of the manual scale and the

data's `fill` values."

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1302	2671	3900	4068	5307	9595

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
3231	6512	11202	14715	19984	95670



2 Genes

```
[10]: gc()

objGenes_SoloTE <- NormalizeData(objGenes_SoloTE, normalization.method = "LogNormalize", scale.factor = 10000)

objGenes_SoloTE <- FindVariableFeatures(objGenes_SoloTE, selection.method = "vst", nfeatures = 4000)

# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objGenes_SoloTE), 10)

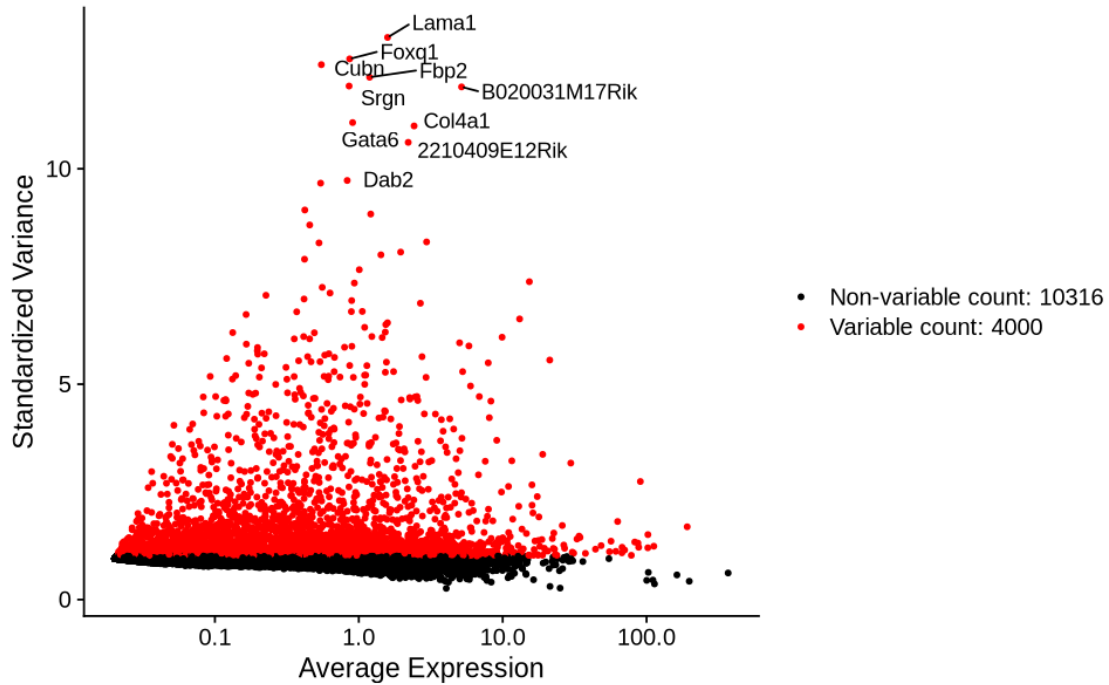
# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objGenes_SoloTE)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2 × 6 of type dbl	Ncells	23302973	1244.6	55531530	2965.8	33729427	1801.4
	Vcells	231319059	1764.9	515269048	3931.2	514867740	3928.2

Normalizing layer: counts

Finding variable features for layer counts

When using repel, set xnudge and ynudge to 0 for optimal results



```
[11]: gc()
all.genes <- rownames(objGenes_SoloTE)
objGenes_SoloTE <- ScaleData(objGenes_SoloTE) # on hvgs

objGenes_SoloTE <- RunPCA(objGenes_SoloTE, features = VariableFeatures(object =
  objGenes_SoloTE))

DimPlot(objGenes_SoloTE, reduction = "pca") + NoLegend()

ElbowPlot(objGenes_SoloTE)
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2 × 6 of type dbl	Ncells	23337397	1246.4	55531530	2965.8	33729427	1801.4
	Vcells	240922120	1838.1	515269048	3931.2	514867740	3928.2

Centering and scaling data matrix

PC_ 1

Positive: Dst, Gm26917, Platr31, Atf2, Rbm25, mt-Nd6, Gm26870, Igfbp2, Gm8300, Gm13228

Scamp1, Nelfa, Gm21358, Gm44739, Os9, Aqr, Drr1, Cacna1s, Gm21761, P4ha2

Ric3, Ercc4, Gm13622, Cwc22, 9530059014Rik, Gm8994, Mg11, B020031M17Rik, Adgrl3, Tmem92

Negative: Rpl32, Pfn1, Eif5a, Hsp90ab1, Rps14, Rps11, Gng5, Rps271, Atp5e, Tceb2

Rpl38, Hmgn1, Actb, Rps5, Rplp1, Ybx1, Prdx1, Hspe1, Rpl6, Cks1b
Atox1, Snrpd1, Snrpe, Atp5j, Cfl1, Rpl37, Eef1a1, Fkbp3, Timm13,

Basp1
PC_ 2

Positive: Gm21761, Platr31, Zfp560, Gm13228, Nelfa, Tsen2, Gm8300, Gm8994, Scamp1, Ercc4

Lonp2, Lgals4, Gm21818, Glrx2, Zfp809, 2810004N23Rik, Sdhaf3,
Gm20767, Mgl1, Aqr

Ppig, Tmem92, Cep57l1, Atp6v1e1, Usp17lc, Gm28940, Spesp1, Cwc22,
Tbc1d12, B020031M17Rik

Negative: Malat1, Kcnq1ot1, Prtg, Eph2, Folr1, Neat1, Meg3, Smad7, Akap12, Tmem37

Gdf3, Cst3, Rlim, Cyr61, Rian, Xbp1, Ckap4, Eph4, Cldn3, Skil
Gm37899, Msmo1, Gja1, Rbms1, Adgrl2, Wee1, Slc7a7, Slc20a1, Mirg,

Pmp22
PC_ 3

Positive: Hoxb1, Hoxb2, Peg10, Mest, Sox4, Igfbp2, Rbp1, Crabp2, Pim1, Id3
Pitx2, Pbx1, Id1, Hotairm1, Lefty1, Hoxb4, Meis2, Id2, Dkc1, Stra8
Hoxb8, Cdkn1c, Psme2, Peg3, Tmsb4x, Uchl1, Grifin, Hoxb5, Hoxb9, Fos

Negative: Fbp2, Srgn, Lama1, Cubn, Gata6, Col4a1, Pdgfra, Col4a2,
2210409E12Rik, Lrpap1

Nostrin, 1010001N08Rik, Kdelr3, Shkbp1, Amn, Foxq1, Dab2, Timd2,
Lamb1, Flrt3

Klb, Gata4, Eif2s2, Gjb3, Htra1, Sox7, Dnmt3b, Gjb5, Sox17, Nrg2

PC_ 4

Positive: Mest, Serpinh1, Sparc, Sox4, Clu, Nostrin, Col4a1, Lama1, Gata6, Cubn
Col4a2, Srgn, Fbp2, Pdgfra, Gja1, Lamb1, Dab2, Prtg, Col1a2, Glipr2
Tshz1, Pmp22, Tpm1, Foxq1, Gata4, F2r, Shkbp1, Amn, Serpine2, Hoxb2

Negative: Gpx4, Chchd10, Dppa5a, Ooep, Jam2, Nanog, Klf2, Spp1, Mylpf, Zfp42
Stmn2, Rif1, Mtf2, Upp1, Fgf4, Klf4, Alg13, Stra8, Csrp2, Ifitm1
Sox2, 1700019D03Rik, Wfdc2, Sept1, Gm7325, Phc1, Prps1, Ticrr, Gsta4,

Tcea3
PC_ 5

Positive: Gm5939, Zfp352, Gm13498, Gm5117, Sp110, A530040E14Rik, Tacr3, Epm2a,
Gm4858, AF067063

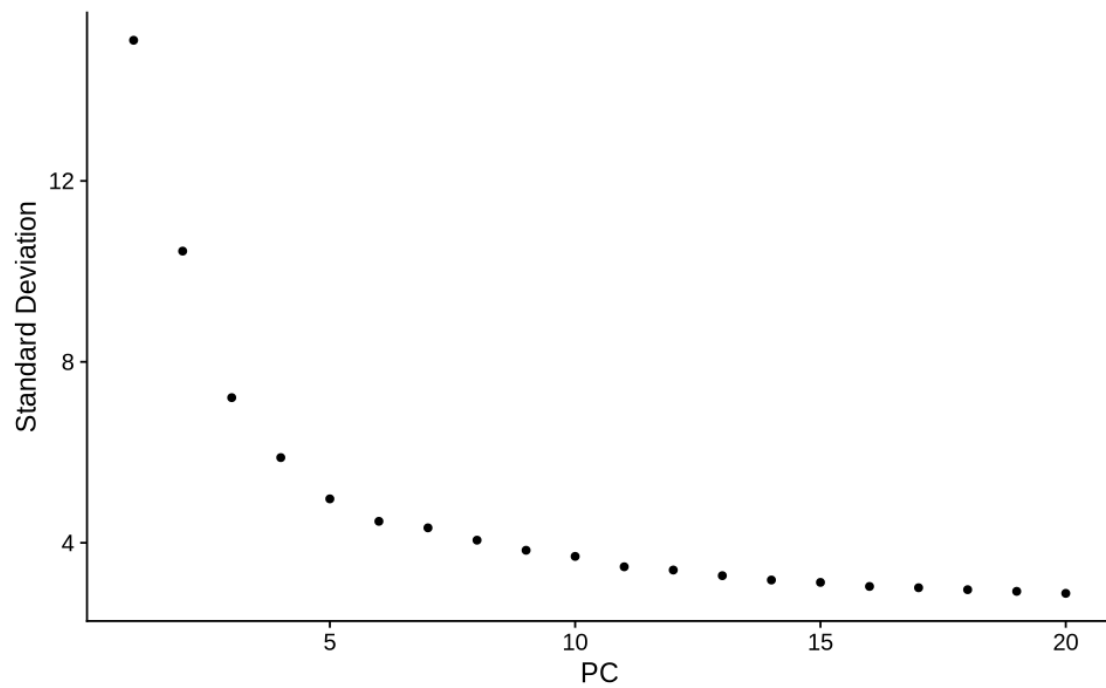
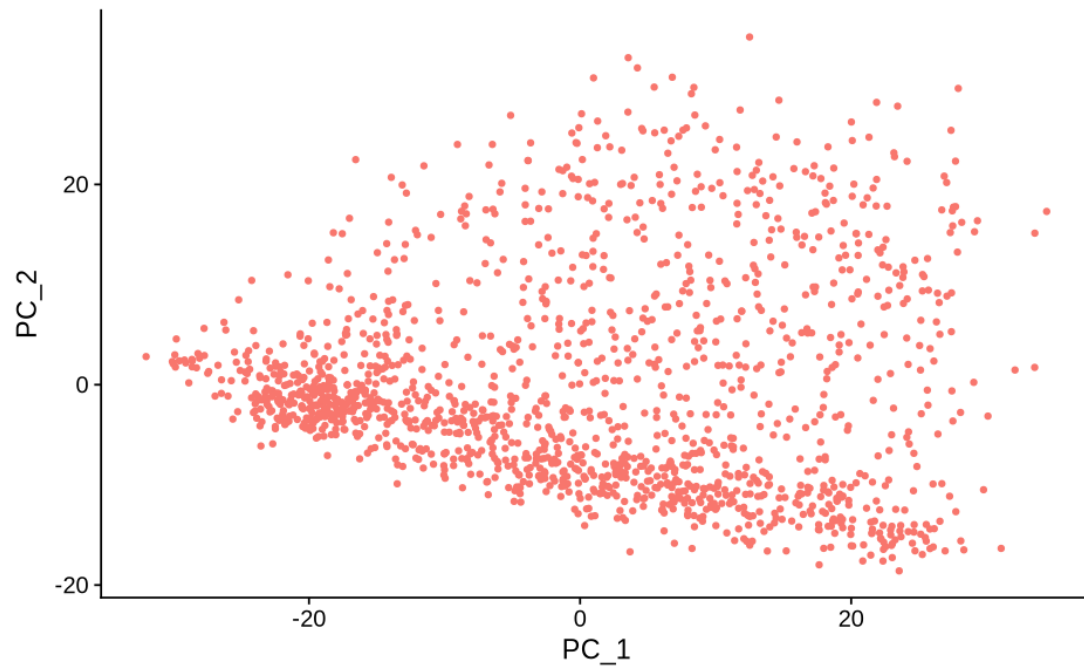
Gm33466, Echdc3, Lmx1a, Gm8953, Rxra, Gm13360, Ttc30b, Usp9y,
Ppp1r14b, Frrs1

Zfp51, Gm16026, Gm7104, L1td1, Csrnp3, Fbxo34, D13Erted608e, BC147527,
B020031M17Rik, Slc16a6

Negative: Avpi1, 3222401L13Rik, Vstm2b, Hotairm1, Calca, Igfbp2, Ctsl,
Tmsb15b2, Hist1h4i, Fgf15

Bdnf, Tmsb15b1, Lgals4, Cox6b2, RP24-560024.2, Hist3h2ba, Ctr9,
Ube2t, Vimp, Them4

Pdpn, Gm12794, Usp17lb, Arl13b, Mtf2, Znhit3, Arl4d, 2210409E12Rik,
Pnp, Areg



```
[12]: objGenes_SoloTE <- FindNeighbors(objGenes_SoloTE, dims = 1:11, k.param = 20)
      objGenes_SoloTE <- FindClusters(objGenes_SoloTE, resolution = 1, algorithm = 4)
```



```
19:11:51 Writing NN index file to temp file
/mnt/TEdataStorage_2T/tmp/RtmpWuPib9/file3d1deb7fe66d30

19:11:51 Searching Annoy index using 1 thread, search_k = 3000

19:11:51 Annoy recall = 100%

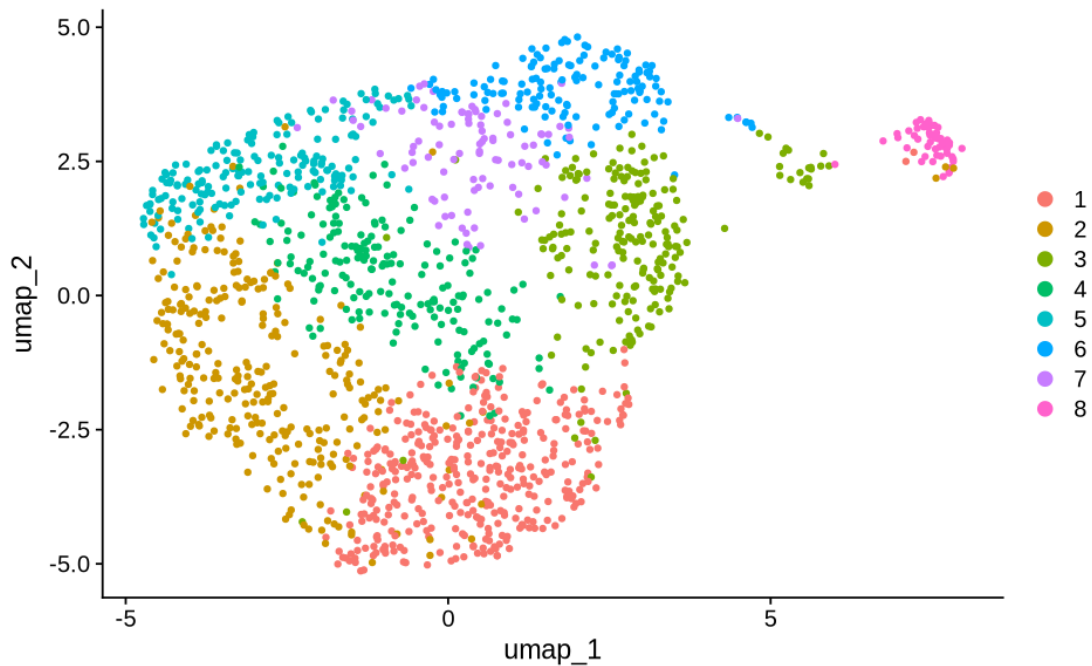
19:11:55 Commencing smooth kNN distance calibration using 1 thread
with target n_neighbors = 30

19:12:03 Initializing from normalized Laplacian + noise (using RSpectra)

19:12:03 Commencing optimization for 500 epochs, with 62222 positive edges
```

19:12:03 Using rng type: pcg

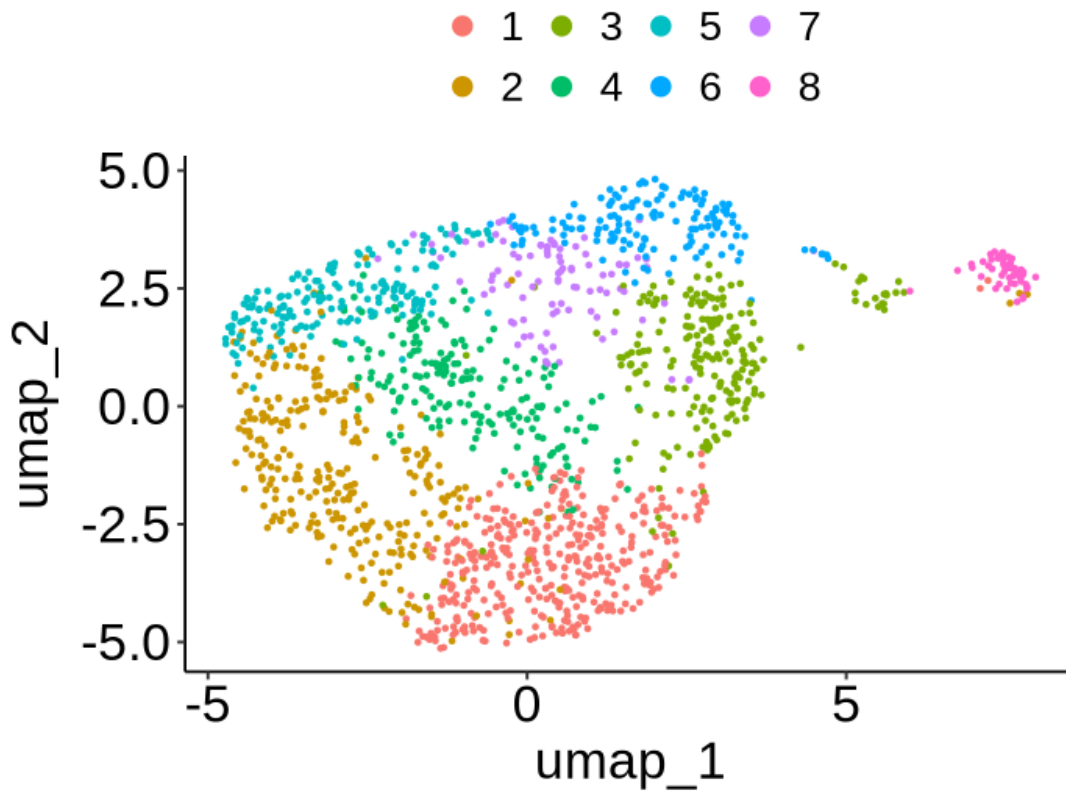
19:12:09 Optimization finished



```
[13]: options(repr.plot.width=6, repr.plot.height=5)

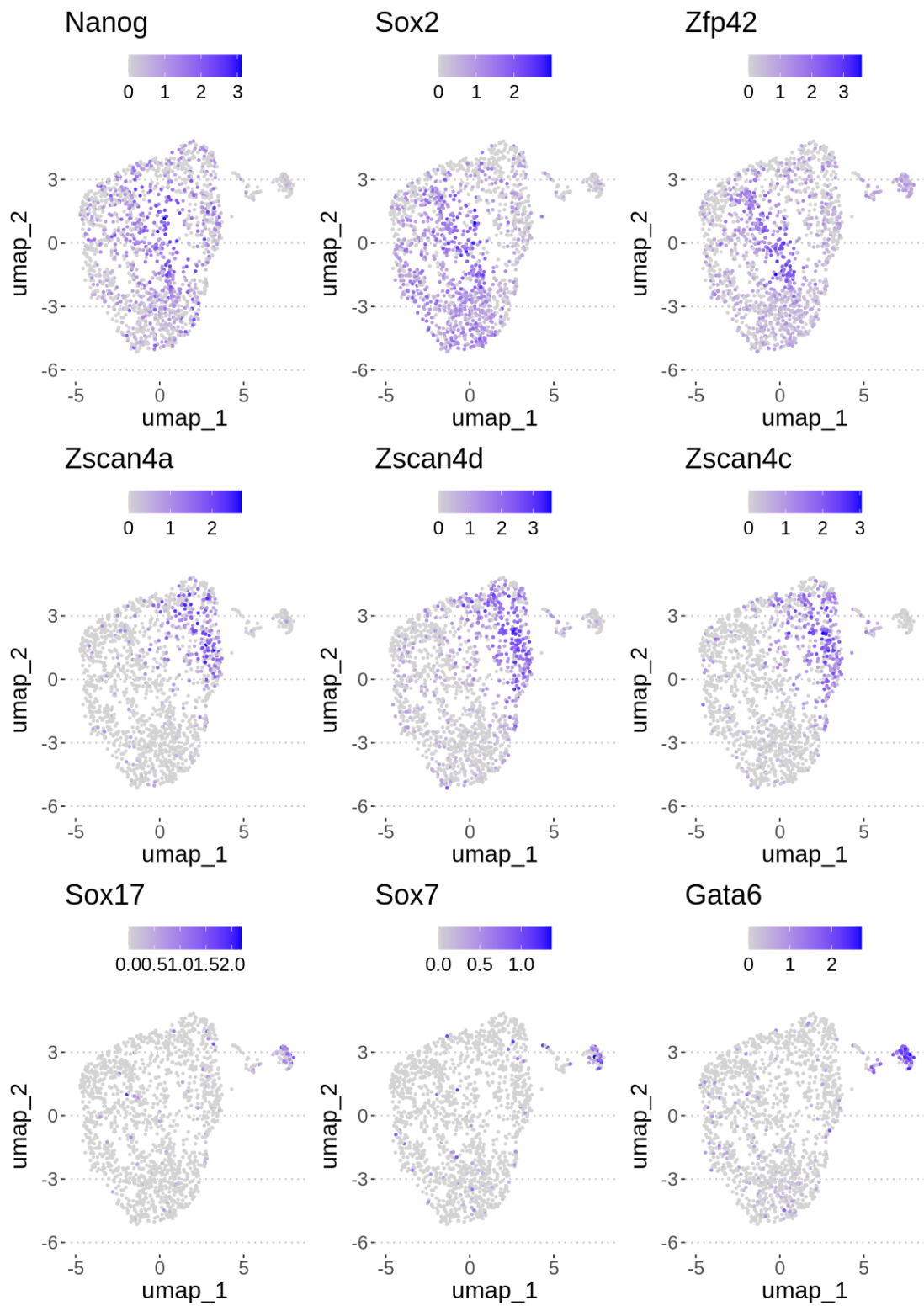
DimPlot(objGenes_SoloTE, reduction = "umap", group.by="RNA_snn_res.1",
        #cols=colorDict,
        shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  theme(text=element_text(size=20))
ggsave("figures_10xMouse_RA/umap_genes_clusters.pdf", device = "pdf", width=6, height=5)
```

RNA_snn_res.1



```
[14]: options(repr.plot.width=10, repr.plot.height=14)

FeaturePlot(objGenes_SoloTE, reduction = "umap", features=c("Nanog", "Sox2",
  ↪ "Zfp42",
  ↪ "Zscan4a", "Zscan4d", "Zscan4c",
  ↪ "Gata6"),
  pt.size = 0.5) &
  theme_pubclean() &
  theme(text=element_text(size=18))
ggsave("figures_10xMouse_RA/umap_genes_markerExpression.pdf", device = "pdf",
  ↪ width=10, height=14)
```




```
[29]: Idents(objGenes_SoloTE) <- as.character(objGenes_SoloTE$`RNA_snn_res.1`)
```

```
objGenes_SoloTE <- RenameIdents(objGenes_SoloTE, '1' = 'Pluripotent')  
objGenes_SoloTE <- RenameIdents(objGenes_SoloTE, '2' = 'Pluripotent')  
objGenes_SoloTE <- RenameIdents(objGenes_SoloTE, '4' = 'Pluripotent')  
objGenes_SoloTE <- RenameIdents(objGenes_SoloTE, '5' = 'Pluripotent')
```

```
objGenes_SoloTE <- RenameIdents(objGenes_SoloTE, '3' = '2CLC')  
objGenes_SoloTE <- RenameIdents(objGenes_SoloTE, '6' = '2CLC')  
objGenes_SoloTE <- RenameIdents(objGenes_SoloTE, '7' = '2CLC')
```

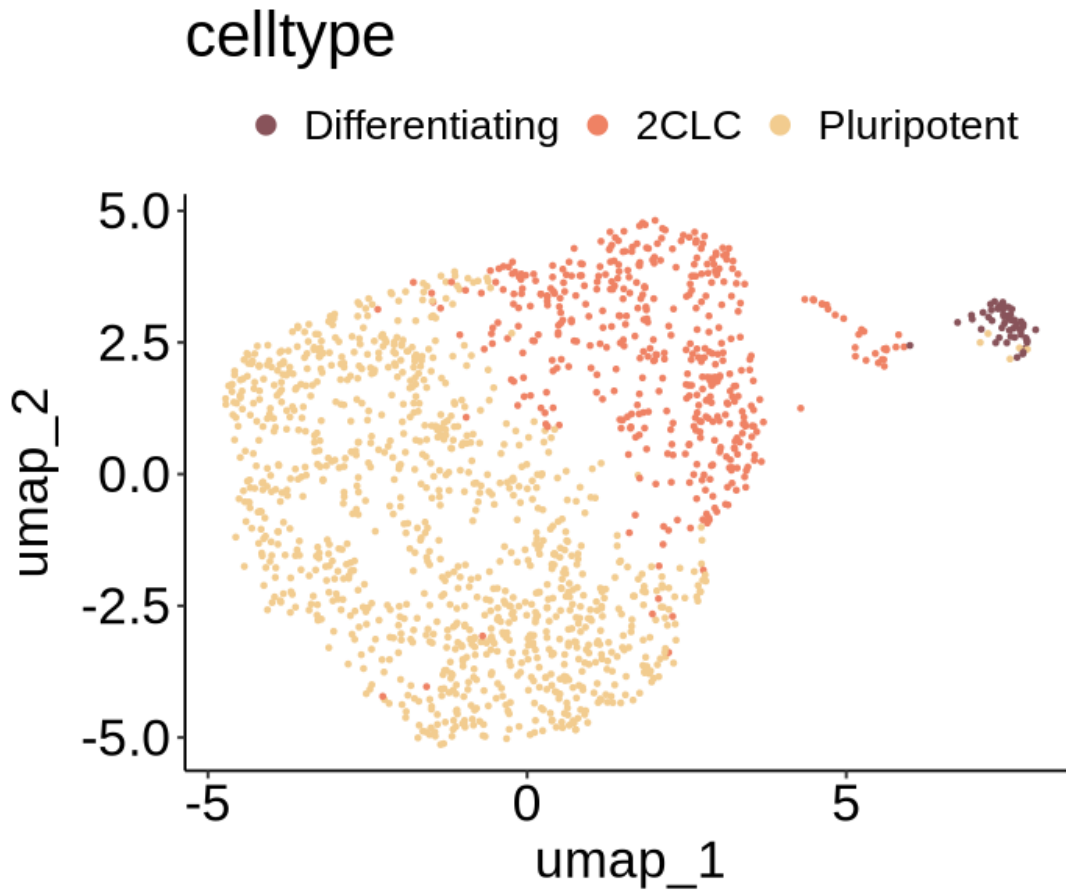
```
objGenes_SoloTE <- RenameIdents(objGenes_SoloTE, '8' = 'Differentiating')
```

```
objGenes_SoloTE$celltype <- Idents(objGenes_SoloTE)
```

```
[31]: options(repr.plot.width=6, repr.plot.height=5)
```

```
DimPlot(objGenes_SoloTE, reduction = "umap", group.by="celltype",  
        cols=colorDict,  
        shuffle=T, pt.size = 0.5) +  
  theme_pubr() +  
  theme(text=element_text(size=20))  
ggsave("figures_10xMouse_RA/umap_genes_celltypes.pdf", device = "pdf")
```

Saving 7 x 7 in image



```
[17]: objGenes_SoloTE <- JoinLayers(objGenes_SoloTE)
```

3 TEs

```
[18]: objTE_SoloTE@assays$RNA
```

Assay (v5) data with 47725 features for 1542 cells

First 10 features:

```
chr10-100308547-100308652-B1-Mus1:Alu:SINE-21.3--,  
chr10-100437762-100439735-Lx7:L1:LINE-20.7--,  
chr10-100484737-100484878-B1-Mus1:Alu:SINE-5.6--,  
chr10-101878262-101879816-Lx7:L1:LINE-26.0--,  
chr10-101880489-101880981-MT2-Mm:ERV1:LTR-1.6--,  
chr10-10243079-10243233-URR1A:hAT-Charlie:DNA-14.2--,  
chr10-103852919-103853310-L1-Mus3:L1:LINE-11.8--,  
chr10-104196602-104198308-MusHAL1:L1:LINE-13.6+,  
chr10-104995219-104995319-L1Md-T:L1:LINE-15.3--,  
chr10-104995320-104995817-MT2-Mm:ERV1:LTR-1.8--
```

Layers:
counts

```
[19]: ### Normalize

objTE_SoloTE <- NormalizeData(objTE_SoloTE, normalization.method = "
  ↪LogNormalize", scale.factor = 10000)

#saveRDS(objTE_SoloTE, file=paste0("data_",dataset_id,"/objTE_SoloTE_beforeHVG.
  ↪RDS"))
```

Normalizing layer: counts

```
[20]: objTE_SoloTE <- FindVariableFeatures(objTE_SoloTE, selection.method = "vst",
  ↪nfeatures = 4000)

# # Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objTE_SoloTE), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_SoloTE)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

Finding variable features for layer counts

When using repel, set xnudge and ynudge to 0 for optimal results

Warning message:

"ggrepel: 5 unlabeled data points (too many overlaps). Consider increasing max.overlaps"

14. 'chr10-4191592-4196918-MERVL-int:ERVL:LTR-1.6-+' 15. 'chr10-44815175-44816126-MERVL-int:ERVL:LTR-1.6-+' 16. 'chr10-50290617-50295942-MERVL-int:ERVL:LTR-1.5-' 17. 'chr10-52206256-52211592-MERVL-int:ERVL:LTR-1.5-' 18. 'chr10-53570238-53575563-MERVL-int:ERVL:LTR-1.4-' 19. 'chr10-62734810-62735867-MERVL-int:ERVL:LTR-1.4-' 20. 'chr10-71013350-71018675-MERVL-int:ERVL:LTR-1.6-' 21. 'chr10-72210358-72211587-MERVL-int:ERVL:LTR-1.6-' 22. 'chr10-79207627-79212067-MERVL-int:ERVL:LTR-1.5-' 23. 'chr10-79884805-79884894-MERVL-int:ERVL:LTR-1.5-' 24. 'chr10-82121367-82122850-MERVL-int:ERVL:LTR-15.2-+' 25. 'chr10-83768851-83769564-MERVL-int:ERVL:LTR-15.2-+' 26. 'chr10-83771391-83774579-MERVL-int:ERVL:LTR-1.9-+' 27. 'chr10-84174998-84180323-MERVL-int:ERVL:LTR-1.9-+' 28. 'chr10-85691270-85691644-MERVL-int:ERVL:LTR-5.1-+' 29. 'chr10-85753206-85755592-MERVL-int:ERVL:LTR-5.1-+' 30. 'chr10-85863490-85868839-MERVL-int:ERVL:LTR-5.3-+' 31. 'chr10-86581763-86584548-MERVL-int:ERVL:LTR-5.3-+' 32. 'chr10-89000402-89000788-MERVL-int:ERVL:LTR-10.5-' 33. 'chr11-104239629-104241880-MERVL-int:ERVL:LTR-1.7-' 34. 'chr11-104241864-104242151-MERVL-int:ERVL:LTR-1.7-' 35. 'chr11-105848504-105853862-MERVL-int:ERVL:LTR-1.7-' 36. 'chr11-119699564-119704883-MERVL-int:ERVL:LTR-1.6-+' 37. 'chr11-16567893-16572390-MERVL-int:ERVL:LTR-1.6-+' 38. 'chr11-19516934-19517752-MERVL-int:ERVL:LTR-0.8-' 39. 'chr11-22396040-22401326-MERVL-int:ERVL:LTR-0.8-' 40. 'chr11-23547019-23552344-MERVL-int:ERVL:LTR-1.3-' 41. 'chr11-26030895-26032254-MERVL-int:ERVL:LTR-1.3-' 42. 'chr11-26032247-26033257-MERVL-int:ERVL:LTR-1.1-' 43. 'chr11-29586480-29591767-MERVL-int:ERVL:LTR-1.1-' 44. 'chr11-40626140-40631465-MERVL-int:ERVL:LTR-1.5-' 45. 'chr11-61548798-61548835-MERVL-int:ERVL:LTR-1.5-' 46. 'chr11-67866840-67872126-MERVL-int:ERVL:LTR-1.6-+' 47. 'chr11-70156341-70161663-MERVL-int:ERVL:LTR-1.6-+' 48. 'chr11-71506364-71511650-MERVL-int:ERVL:LTR-1.4-' 49. 'chr11-72903324-72903797-MERVL-2A-int:ERVL:LTR-1.4-' 50. 'chr11-7586034-7591357-MERVL-int:ERVL:LTR-1.5-+'

```
[22]: objTE_SoloTE <- RunPCA(objTE_SoloTE, features = VariableFeatures(object = objTE_SoloTE))

DimPlot(objTE_SoloTE, reduction = "pca") + NoLegend()

ElbowPlot(objTE_SoloTE)
```

PC_ 1

Positive: chr15-38053642-38054092-Charlie1:hAT-Charlie:DNA-32.3--,
chr19-5830694-5831063-L1MB7:L1:LINE-29.3-+,
chrX-36861582-36861681-PB1D9:Alu:SINE-22.0-+,
chrX-152254939-152255127-B2-Mm2:B2:SINE-4.3--,
chr10-77162792-77162938-B1-Mus1:Alu:SINE-6.1-+,
chr1-135745659-135745943-B4A:B4:SINE-26.7-+,
chr1-8937253-8937403-B1-Mus2:Alu:SINE-4.2-+,
chr18-77057482-77057683-ORR1D2:ERVL-MaLR:LTR-20.6--,
chr15-38053498-38053641-B1-Mus2:Alu:SINE-5.6--,
chr9-6199563-6204202-L1VL4:L1:LINE-14.1-+
chr1-133016852-133017015-B2-Mm1a:B2:SINE-7.9--,
chr10-117698919-117699030-B1-Mur1:Alu:SINE-17.8-+,
chrX-94135864-94135918-ORR1D1:ERVL-MaLR:LTR-16.4-+,
chr15-75086463-75086531-B1-Mur4:Alu:SINE-24.6-+,
chr5-53573469-53573579-L1MB7:L1:LINE-23.6--,
chr7-44822084-44822272-B2-Mm2:B2:SINE-14.2--,
chr11-117045307-117045481-B2-Mm2:B2:SINE-6.3--,
chr4-132531537-132531648-B1-Mur2:Alu:SINE-18.4--,
chr17-70924651-70924786-RSINE1:B4:SINE-26.5-+ , chr8-12476783-12477182-MTC:ERVL-

MaLR:LTR-12.8-+

chr12-71200599-71201021-L1-Rod:L1:LINE-26.6--,
chr1-8936520-8937252-MMERVK9C-I-int:ERVK:LTR-20.1-+,
chr3-121430258-121430449-B2-Mm1t:B2:SINE-12.0--,
chr7-26694384-26700082-ETnERV2-int:ERVK:LTR-13.4--,
chr15-36585488-36585619-B1-Mur4:Alu:SINE-18.9--,
chrX-139324342-139326244-L1MA6:L1:LINE-30.2-+,
chr6-5334581-5337260-ETnERV2-int:ERVK:LTR-16.3-+,
chr3-9414280-9415836-L1-Mur3:L1:LINE-16.6--,
chr17-34125652-34125719-ID-B1:B4:SINE-23.5-+,
chr3-9380474-9386500-Lx:L1:LINE-9.2--
Negative: chr5-108190967-108191071-RSINE1:B4:SINE-27.2-+,
chr10-50284606-50286633-Lx6:L1:LINE-25.3--,
chr8-86919477-86919656-B2-Mm2:B2:SINE-10.6-+,
chr3-140100191-140100580-Lx4A:L1:LINE-16.6--,
chrX-109293101-109299056-Lx2B:L1:LINE-15.2-+,
chr4-77631440-77631636-B3:B2:SINE-35.6-+,
chr2-142473338-142473442-B4:B4:SINE-29.5-+, chr17-23521272-23521679-ERVB7-4-LTR-MM:ERVK:LTR-9.3-+, chrX-126775095-126775185-L1Md-F2:L1:LINE-11.5-+,
chr11-54099073-54099689-Lx3C:L1:LINE-11.3-+

chr17-23504906-23510225-MERVL-int:ERVL:LTR-1.5-+,
chr1-16910701-16911254-RMER17C-int:ERVK:LTR-19.2--,
chr1-13596843-13597144-ORR1A2:ERVL-MaLR:LTR-5.6-+,
chr2-113102443-113102542-L2b:L2:LINE-38.0-+,
chr15-6545853-6546345-MT2-Mm:ERVL:LTR-0.4-- , chr12-12634270-12634591-MTC-int:ERVL-MaLR:LTR-15.9-- , chr7-97114148-97114240-ID-B1:B4:SINE-29.0-+ ,
chr17-23520084-23521271-L1-Mus3:L1:LINE-11.9-+ ,
chr7-69246657-69246705-B1-Mur3:Alu:SINE-6.3-+ ,
chr13-106533748-106533888-B1-Mus1:Alu:SINE-10.6-+

chr6-68131695-68133852-L1-Mur2:L1:LINE-15.7-- ,
chr12-12634123-12634269-B1-Mus1:Alu:SINE-8.8-- ,
chr1-26310708-26315921-L1VL4:L1:LINE-12.5-+ ,
chr14-28425590-28426079-L1-Mus1:L1:LINE-6.8-- ,
chrX-78947736-78948251-ERVB4-2-LTR-MM:ERVK:LTR-7.6-+ ,
chr12-56103446-56103776-IAPEY3-LTR:ERVK:LTR-3.0-+ ,
chrX-150305393-150305867-MT2B2:ERVL:LTR-12.4-- ,
chr17-20719990-20721846-Lx3-Mus:L1:LINE-11.7-+ ,
chr1-8153313-8154968-RMER16-int:ERVK:LTR-11.4-+ ,
chr11-9815048-9815409-L1Md-F2:L1:LINE-7.3-+

PC_ 2

Positive: chr11-43906182-43907454-L1-Mur3:L1:LINE-31.2-+ ,
chr10-61032796-61032889-B1F:Alu:SINE-35.1-- ,
chr13-84654494-84655725-L1-Mur1:L1:LINE-10.3-- ,
chr11-43908181-43908893-L1-Mur3:L1:LINE-26.9-+ ,
chr13-84655979-84657928-L1-Mur1:L1:LINE-10.3-- , chr12-12629713-12629869-MTB-Mm:ERVL-MaLR:LTR-14.5-+ , chr10-61032846-61032978-ID-B1:B4:SINE-30.8-- ,
chr11-43907999-43908180-B3:B2:SINE-21.7-+ ,
chr3-119611520-119612233-Lx3-Mus:L1:LINE-9.7-- ,

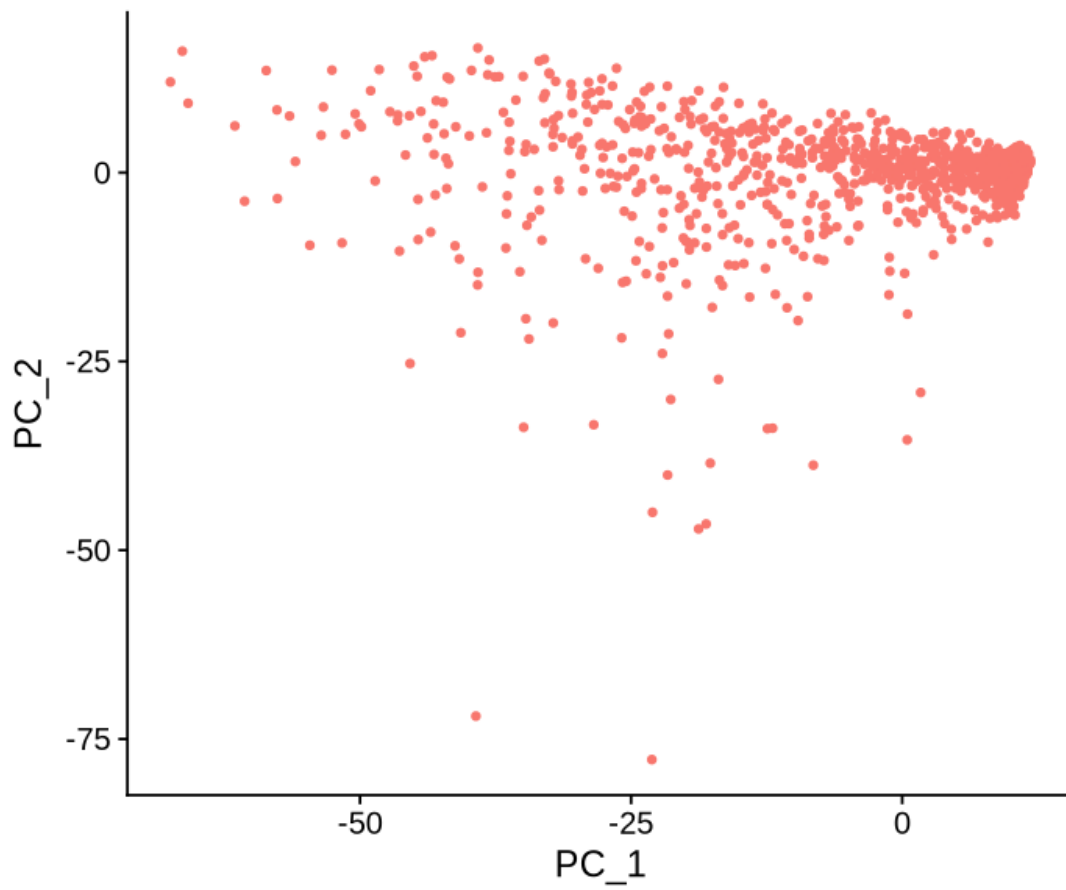
chr9-119988764-119988941-B2-Mm1t:B2:SINE-8.5-+
 chrX-94063317-94063505-B3A:B2:SINE-25.9-+,
 chr12-12614238-12614557-Charlie25:hAT-Charlie:DNA-31.1-+,
 chr11-25931943-25934781-L1VL4:L1:LINE-11.2-- , chr12-12601429-12601827-MTD:ERVL-
 MaLR:LTR-23.8-- , chr10-61032661-61032777-PB1D9:Alu:SINE-24.8-- ,
 chr12-12634123-12634269-B1-Mus1:Alu:SINE-8.8-- ,
 chr8-3551900-3552176-ETnERV2-int:ERVK:LTR-28.3-- ,
 chr11-43909114-43909280-ORR1D1:ERVL-MaLR:LTR-22.2-- ,
 chr5-51355259-51355527-MTE2b:ERVL-MaLR:LTR-37.1-- ,
 chr13-84657964-84660266-L1Md-F2:L1:LINE-8.2--
 chr13-84653551-84653866-Lx3A:L1:LINE-16.0-- ,
 chr2-27791551-27791611-MT2B:ERVL:LTR-8.2-- ,
 chr11-16577138-16577969-L1-Mus2:L1:LINE-7.1-+ ,
 chr2-27791646-27791975-MLTR14:ERV1:LTR-23.8-+ ,
 chr8-52236291-52237932-L1VL2:L1:LINE-16.4-+ ,
 chr10-36868861-36871262-L1-Mus3:L1:LINE-11.5-- ,
 chr3-60905093-60905564-L1M2:L1:LINE-30.8-+ ,
 chr9-30703919-30704909-L1-Mur3:L1:LINE-26.2-+ ,
 chr2-40161998-40162153-Lx3A:L1:LINE-11.1-- ,
 chr3-93766883-93767024-RMER17C:ERVK:LTR-25.2--
 Negative: chr2-181558871-181559029-B2-Mm1a:B2:SINE-8.5-- ,
 chr8-36636100-36636287-B2-Mm2:B2:SINE-9.0-- ,
 chr16-33797648-33797756-B1-Mus1:Alu:SINE-5.5-- ,
 chrX-106017738-106017918-B2-Mm1t:B2:SINE-10.5-+ ,
 chr2-30231841-30232037-B2-Mm2:B2:SINE-9.3-- ,
 chr11-75347278-75347414-RSINE1:B4:SINE-27.1-+ ,
 chr16-33797659-33797768-ID-B1:B4:SINE-18.7-- ,
 chr8-106554538-106554696-B2-Mm1a:B2:SINE-3.1-- ,
 chr16-33797786-33797978-B2-Mm1a:B2:SINE-3.1-- ,
 chr7-39527243-39527430-B2-Mm1t:B2:SINE-6.9--
 chr4-108321449-108321639-B2-Mm1a:B2:SINE-8.4-+ ,
 chr3-95351833-95352023-B2-Mm1a:B2:SINE-8.9-+ ,
 chr17-78711797-78711980-B2-Mm1t:B2:SINE-7.3-+ ,
 chr16-18848586-18848770-B2-Mm1a:B2:SINE-7.0-- ,
 chr7-12722112-12722303-B2-Mm1a:B2:SINE-2.6-+ ,
 chr14-120771913-120772972-RMER16-int:ERVK:LTR-11.9-- ,
 chr5-18514960-18515152-B2-Mm1a:B2:SINE-1.6-+ ,
 chr4-121113403-121113578-B2-Mm1t:B2:SINE-9.1-- ,
 chr11-100303654-100303828-B2-Mm2:B2:SINE-7.4-- ,
 chr9-65895919-65896107-B2-Mm2:B2:SINE-9.5--
 chr2-91752352-91752545-B2-Mm1a:B2:SINE-4.1-- ,
 chr5-30041007-30041195-B2-Mm1t:B2:SINE-2.6-- ,
 chr5-134343037-134343213-B2-Mm2:B2:SINE-7.9-+ ,
 chr16-18848329-18848524-B2-Mm1a:B2:SINE-7.3-- ,
 chr13-55627206-55627383-B3:B2:SINE-18.7-- ,
 chr12-59039209-59039403-B2-Mm1t:B2:SINE-10.1-- ,
 chr2-181496061-181496251-B2-Mm1t:B2:SINE-2.1-+ ,
 chr11-95288121-95288307-B2-Mm2:B2:SINE-7.5-+ ,

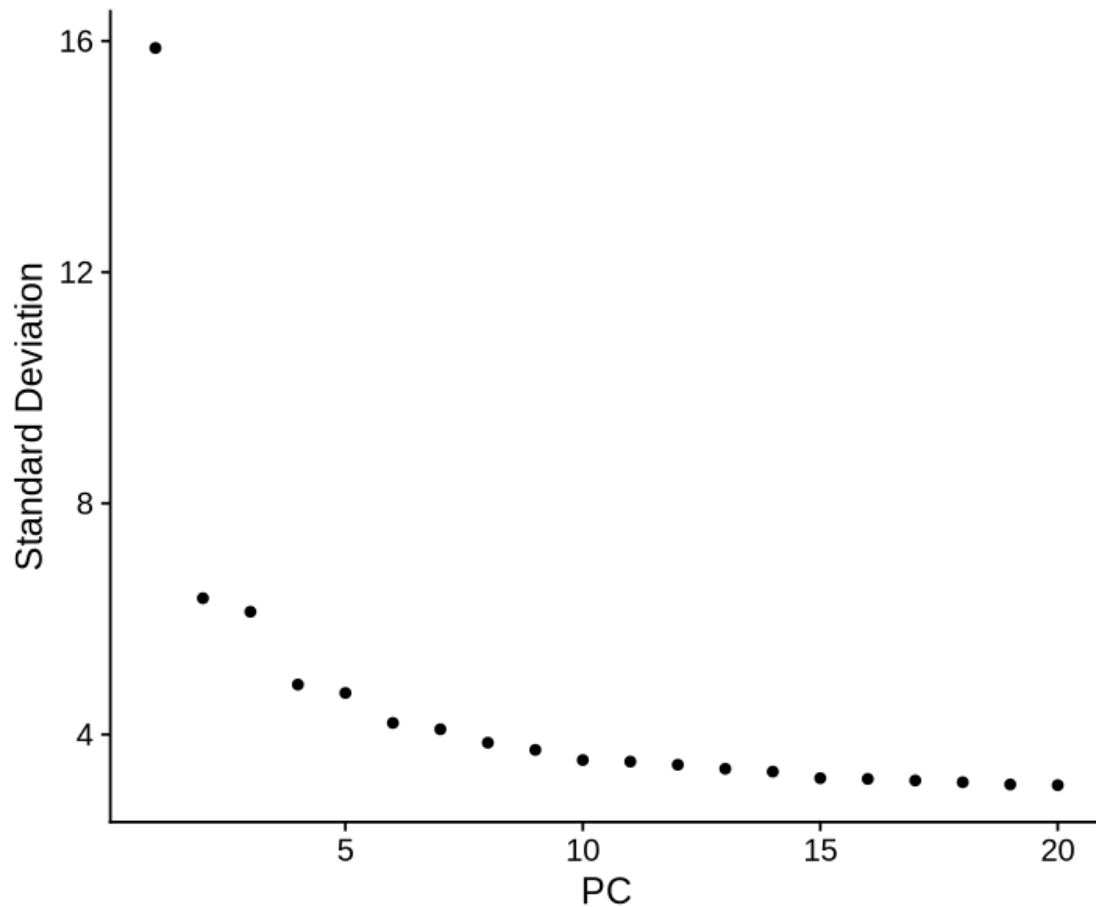
chr9-103200976-103201163-B3A:B2:SINE-26.2-+,
chr13-21267605-21267778-B2-Mm2:B2:SINE-16.3--
PC_ 3
Positive: chr19-8077073-8082738-Lx2A:L1:LINE-15.0-+,
chr19-8085987-8088363-L1Md-F2:L1:LINE-17.3-+,
chr19-8073446-8074138-Lx2B:L1:LINE-17.1--,
chr19-8083154-8084853-L1-Mur2:L1:LINE-16.3-+,
chr3-94156985-94157174-B3:B2:SINE-25.5--,
chr3-93766883-93767024-RMER17C:ERVK:LTR-25.2--,
chr17-44765457-44765541-MTC:ERVL-MaLR:LTR-16.7-+,
chr17-44765546-44765686-B1-Mus1:Alu:SINE-6.4-+,
chr3-93767025-93767172-B1-Mus1:Alu:SINE-13.0-+,
chr5-143192602-143192733-RSINE1:B4:SINE-30.3-+
chr14-47450999-47451146-B1-Mus1:Alu:SINE-6.1--,
chr8-31736868-31736995-ID-B1:B4:SINE-25.6--,
chr14-47450837-47450972-B1-Mm:Alu:SINE-6.2--,
chr17-44764963-44765456-MT2-Mm:ERVL:LTR-4.1-+,
chr3-134901975-134902384-L1Md-F2:L1:LINE-6.2-+,
chr14-47450796-47450808-B1-Mm:Alu:SINE-6.2--,
chr12-66064141-66064649-MT2-Mm:ERVL:LTR-7.7-+,
chr5-51355259-51355527-MTE2b:ERVL-MaLR:LTR-37.1--,
chr14-47455865-47456044-ID-B1:B4:SINE-26.7-+,
chr12-66065603-66070644-L1-Mus3:L1:LINE-8.9-+
chr6-9858265-9858407-B1-Mus2:Alu:SINE-9.8--,
chr14-43844637-43844801-L1Md-F2:L1:LINE-4.2--,
chr3-93633277-93633610-RMER17C:ERVK:LTR-16.0-+,
chrX-117267281-117267471-B3:B2:SINE-31.2-+,
chr6-9858408-9858765-L1-Mus3:L1:LINE-9.3--,
chr14-43844802-43845294-MT2-Mm:ERVL:LTR-1.4--,
chr3-93797933-93798034-Lx3A:L1:LINE-11.8-+,
chr3-52437530-52437592-Lx9:L1:LINE-24.2--,
chr3-93872722-93872852-B3A:B2:SINE-33.6--,
chr6-148646183-148647151-Lx5c:L1:LINE-16.8-+
Negative: chr19-9909916-9910055-L1MB3:L1:LINE-23.7-+,
chr15-102607605-102607740-B1F:Alu:SINE-26.7--,
chr19-9910078-9910230-B1-Mus2:Alu:SINE-4.8-+,
chr10-58767528-58771781-L1Md-F3:L1:LINE-4.9-+,
chr1-140722573-140727921-L1Md-F3:L1:LINE-7.3-+,
chr15-102607903-102608020-B1F2:Alu:SINE-22.5--,
chr19-9910232-9910787-LTRIS-Mus:ERV1:LTR-5.4-+,
chr11-94816700-94816843-B1-Mus2:Alu:SINE-10.4--,
chr19-23264733-23265056-ORR1A4:ERVL-MaLR:LTR-13.2-+,
chr11-54099073-54099689-Lx3C:L1:LINE-11.3-+
chr15-102607889-102607902-ID-B1:B4:SINE-31.2--,
chr11-40571797-40571971-B2-Mm2:B2:SINE-18.1-+,
chr14-75841707-75841899-MTA-Mm:ERVL-MaLR:LTR-1.0-+,
chr5-13709574-13710027-MT2A:ERVL:LTR-29.0--,
chr5-26071907-26072170-MMERVK10C-int:ERVK:LTR-7.6-+,
chr15-14768164-14768843-Lx4A:L1:LINE-18.5--,
chr18-80030122-80030199-X7C-LINE:CR1:LINE-30.7-+,

chr18-80030263-80030464-Lx3B:L1:LINE-22.4-+,
 chr19-23263927-23264419-MT2-Mm:ERVL:LTR-1.0-+,
 chr8-86919477-86919656-B2-Mm2:B2:SINE-10.6-+
 chr19-23265304-23265439-ID-B1:B4:SINE-25.5--,
 chr19-9913511-9913618-L1MA5:L1:LINE-25.2--,
 chr15-82996722-82997031-Lx5b:L1:LINE-16.1--,
 chr4-35175730-35175918-RMER17C2:ERVK:LTR-23.7-+,
 chr5-138074792-138075025-L1MA9:L1:LINE-29.9-+,
 chr1-13596843-13597144-ORR1A2:ERVL-MaLR:LTR-5.6-+,
 chr15-14759966-14761952-Lx3B:L1:LINE-14.0--,
 chr3-93821224-93821700-RMER17C2:ERVK:LTR-24.8--,
 chr19-34140020-34145048-L1-Mus1:L1:LINE-7.2--,
 chr7-45577245-45578018-RLTR13G:ERVK:LTR-12.6--
 PC_ 4
 Positive: chr18-84936343-84936476-MLT1B:ERVL-MaLR:LTR-34.3-+,
 chr18-84936495-84936629-L1MC4:L1:LINE-28.5--,
 chr18-84939954-84940099-B1-Mus2:Alu:SINE-7.5--,
 chr9-119988764-119988941-B2-Mm1t:B2:SINE-8.5-+,
 chr9-119988657-119988755-B1-Mus2:Alu:SINE-7.1-+,
 chr9-108650859-108650970-B1-Mus2:Alu:SINE-9.1-+,
 chr12-118914362-118914474-B3:B2:SINE-29.6--,
 chr16-13154920-13155112-B3:B2:SINE-20.7-+, chr1-13605811-13606189-ORR1F:ERVL-
 MaLR:LTR-29.4-+, chr17-35601392-35601582-MTD:ERVL-MaLR:LTR-28.8-+
 chr15-76945218-76945319-MTC-int:ERVL-MaLR:LTR-18.0--,
 chr15-59292810-59293472-ORR1A2-int:ERVL-MaLR:LTR-10.1-+,
 chr12-118914230-118914361-B1-Mus1:Alu:SINE-19.7--,
 chr15-83008919-83008947-PB1D11:Alu:SINE-23.6-+,
 chr2-171795167-171795335-ID-B1:B4:SINE-29.4-+,
 chr15-76945081-76945217-B1-Mm:Alu:SINE-5.8--,
 chr4-53300678-53301960-L1-Mur2:L1:LINE-13.3--,
 chr16-8928652-8928877-L1M5:L1:LINE-35.0-+,
 chr9-119985860-119985942-ID4:ID:SINE-20.7--,
 chr6-49436162-49437261-L1-Mur3:L1:LINE-15.5--
 chr14-47462179-47462356-L1MC1:L1:LINE-32.5-+,
 chr12-21448133-21448260-B1-Mus2:Alu:SINE-8.6--,
 chr1-134954340-134954490-ID-B1:B4:SINE-21.2-+,
 chr8-90800629-90800823-B3:B2:SINE-18.9--,
 chr11-17256904-17257362-Lx2B2:L1:LINE-16.5-+,
 chr9-119988058-119988197-B1-Mus2:Alu:SINE-4.3-+,
 chr18-84935292-84935443-RMER17B:ERVK:LTR-14.3--,
 chr2-171818037-171818528-MT2-Mm:ERVL:LTR-0.4--,
 chr9-19093797-19094158-MMERVK9C-I-int:ERVK:LTR-15.6-+,
 chr6-85665245-85665410-B2-Mm1t:B2:SINE-9.6--
 Negative: chr12-17859704-17859802-RLTR10B2:ERVK:LTR-12.1-+,
 chr9-117037470-117037643-L1M3:L1:LINE-28.8-+,
 chr1-30619584-30620272-Lx6:L1:LINE-24.9--,
 chr4-3382933-3383007-B3A:B2:SINE-22.7-+, chr19-23264733-23265056-ORR1A4:ERVL-
 MaLR:LTR-13.2-+, chr10-52304855-52305060-B3:B2:SINE-23.0-+,

chr11-40571797-40571971-B2-Mm2:B2:SINE-18.1-+,
 chr12-17859400-17859700-IAPEY3C-LTR:ERVK:LTR-1.7-+,
 chr13-35565109-35565580-MT2C-Mm:ERVL:LTR-12.8--,
 chr15-6545853-6546345-MT2-Mm:ERVL:LTR-0.4--
 chr8-37470076-37470568-MT2-Mm:ERVL:LTR-1.6-+,
 chr4-88651116-88651634-L1-Mur3:L1:LINE-27.1-+,
 chr10-72815017-72815131-Lx8:L1:LINE-25.5-+,
 chr6-42032589-42033412-Lx2B2:L1:LINE-18.6-+, chr2-10172121-10172428-ORR1A2:ERVL-
 MaLR:LTR-9.4-- , chr1-8153313-8154968-RMER16-int:ERVK:LTR-11.4-+,
 chr8-37478122-37478248-MTE2b-int:ERVL-MaLR:LTR-28.7-+,
 chr12-17859803-17859888-MMERVK10C-int:ERVK:LTR-7.0-+,
 chr14-85319421-85319635-L1-Mus3:L1:LINE-16.3-+,
 chr1-80388189-80388675-MT2-Mm:ERVL:LTR-2.3--
 chr19-23265304-23265439-ID-B1:B4:SINE-25.5-- ,
 chr2-80232584-80233083-MLT2B1:ERVL:LTR-30.2-- ,
 chr1-30620314-30620803-L1Md-F2:L1:LINE-5.1-- ,
 chr18-53859086-53859678-MT2B2:ERVL:LTR-9.3-+,
 chr16-85936758-85937404-RMER20B:ERVK:LTR-18.9-+,
 chr12-84089925-84090227-IAPEY3-LTR:ERVK:LTR-11.9-- ,
 chr7-122962405-122962897-MT2-Mm:ERVL:LTR-1.0-- ,
 chr7-138922026-138922166-B1-Mus2:Alu:SINE-13.5-- ,
 chr2-167520523-167521015-MT2-Mm:ERVL:LTR-1.6-+,
 chr7-122961319-122961415-PB1D7:Alu:SINE-26.0--
 PC_ 5
 Positive: chr3-95431228-95431317-B4:B4:SINE-35.7-- ,
 chr3-95431131-95431202-B4:B4:SINE-35.7-- ,
 chr7-43179388-43179768-Lx7:L1:LINE-17.1-- ,
 chr9-22159858-22160321-MT2C-Mm:ERVL:LTR-4.3-- ,
 chr9-22159461-22159570-B1-Mus1:Alu:SINE-3.8-- ,
 chr3-95431365-95431856-MT2-Mm:ERVL:LTR-0.6-- ,
 chr2-157242709-157243201-MT2-Mm:ERVL:LTR-1.4-+,
 chr17-63589946-63590242-L1Md-F2:L1:LINE-4.2-+,
 chr12-17866178-17866670-MT2-Mm:ERVL:LTR-2.0-- ,
 chr2-128955316-128955456-B1-Mur3:Alu:SINE-21.2-+
 chr12-88324440-88324504-B2-Mm2:B2:SINE-16.0-+,
 chr16-3994056-3994204-B1-Mus1:Alu:SINE-4.1-+,
 chr9-22159594-22159740-B1-Mus2:Alu:SINE-8.2-- ,
 chr7-43179769-43180261-MT2-Mm:ERVL:LTR-1.6-- ,
 chr7-43179189-43179334-B1-Mm:Alu:SINE-8.2-- ,
 chr9-22233686-22233982-RLTR48A:ERV1:LTR-14.4-+,
 chr2-128955176-128955315-L1MD:L1:LINE-23.4-+,
 chr4-147306565-147307052-MT2-Mm:ERVL:LTR-1.4-+,
 chr8-69408931-69414085-L1-Mus1:L1:LINE-7.7-+,
 chr12-21587459-21587805-ORR1A1:ERVL-MaLR:LTR-5.8-+
 chr7-140429326-140432279-L1Md-F2:L1:LINE-6.3-- ,
 chr1-157460148-157460258-PB1D9:Alu:SINE-22.7-- ,
 chr16-13914772-13914935-B3:B2:SINE-17.4-+,
 chr13-120172822-120172920-B1-Mur3:Alu:SINE-21.2-- ,

chr11-29111868-29113381-L1-Mus1:L1:LINE-7.9-+,
chr17-82994903-82995097-L1MC4:L1:LINE-31.4-+,
chr7-73686847-73687356-MT2C-Mm:ERVL:LTR-3.6-+,
chr9-108601812-108601882-B3:B2:SINE-15.7-+,
chr8-73588223-73592639-Lx:L1:LINE-11.7--,
chr16-13914279-13914771-MT2-Mm:ERVL:LTR-0.8-+
Negative: chr1-26310708-26315921-L1VL4:L1:LINE-12.5-+,
chrX-67505899-67506757-MTE-int:ERVL-MaLR:LTR-30.2-+,
chrX-67508149-67508313-L1-Mus1:L1:LINE-7.9-+,
chr10-50241165-50241691-L1-Rod:L1:LINE-25.5--,
chr1-98816140-98817404-MLTR25C:ERVK:LTR-18.0-+,
chrX-15920985-15921196-L1MED:L1:LINE-27.0--,
chr16-54278847-54279813-L1-Mur2:L1:LINE-17.0--,
chr8-14308115-14309144-L1MC2:L1:LINE-27.1-+, chr8-16856940-16857204-MLT1A0:ERVL-
MaLR:LTR-29.7-+, chr3-60927805-60928151-ORR1E:ERVL-MaLR:LTR-34.3-+
chr1-113791699-113794018-L1Md-F2:L1:LINE-12.4-+,
chr3-60927593-60927739-B1-Mus1:Alu:SINE-5.4--,
chr18-45515266-45517152-Lx3C:L1:LINE-11.8-+,
chrX-95584167-95589350-L1Md-F2:L1:LINE-7.3-+,
chrX-67507881-67508148-Lx8b:L1:LINE-24.5-+,
chr8-17041023-17041211-B2-Mm2:B2:SINE-12.7--,
chr2-140312028-140319193-L1-Mus1:L1:LINE-14.7--,
chr3-112882355-112886607-L1-Mus3:L1:LINE-10.0--,
chr18-19329951-19331075-Lx3-Mus:L1:LINE-10.7--,
chr9-117037470-117037643-L1M3:L1:LINE-28.8-+
chr15-21789313-21791870-Lx3C:L1:LINE-17.3--,
chr12-12606246-12606659-Lx3B:L1:LINE-11.7-+,
chr3-134039769-134042709-L1-Mur3:L1:LINE-30.4-+,
chr8-29245300-29245454-B3A:B2:SINE-26.4--,
chr1-16926219-16926447-Lx2:L1:LINE-12.4-+,
chrX-67500898-67501333-RLTR22-Mur:ERVK:LTR-10.9-+,
chr8-52199913-52200445-Lx9:L1:LINE-28.0-- , chr8-16905036-16905239-URR1B:hAT-
Charlie:DNA-19.8-- , chr16-47502294-47502766-Lx10:L1:LINE-24.9-- ,
chr5-70515676-70521645-L1Md-F2:L1:LINE-8.6--





```
[23]: objTE_SoloTE <- FindNeighbors(objTE_SoloTE, dims = 1:12, k.param = 20)
      objTE_SoloTE <- FindClusters(objTE_SoloTE, resolution = 1, algorithm=4)
      objTE_SoloTE <- RunUMAP(objTE_SoloTE, dims = 1:12)
      DimPlot(objTE_SoloTE, reduction = "umap")
```

Computing nearest neighbor graph

Computing SNN

Warning message in RunLeiden(object = object, method = method, partition.type = "RBConfigurationVertexPartition", :

"`random.seed` must be greater than 0 for leiden clustering, resetting
`random.seed` to 1."

19:38:33 UMAP embedding parameters a = 0.9922 b = 1.112

19:38:33 Read 1542 rows and found 12 numeric columns

19:38:33 Using Annoy for neighbor search, n_neighbors = 30

```
19:38:33 Building Annoy index with metric = cosine, n_trees = 50
```

0% 10 20 30 40 50 60 70 80 90 100%

[-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

*
*
*
*
*
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*
|

19:38:33 Writing NN index file to temp file
/mnt/TEdataStorage_2T/tmp/RtmpWuPib9/file3d1deb4d554fa5

19:38:33 Searching Annoy index using 1 thread, search_k = 3000

19:38:33 Annoy recall = 100%

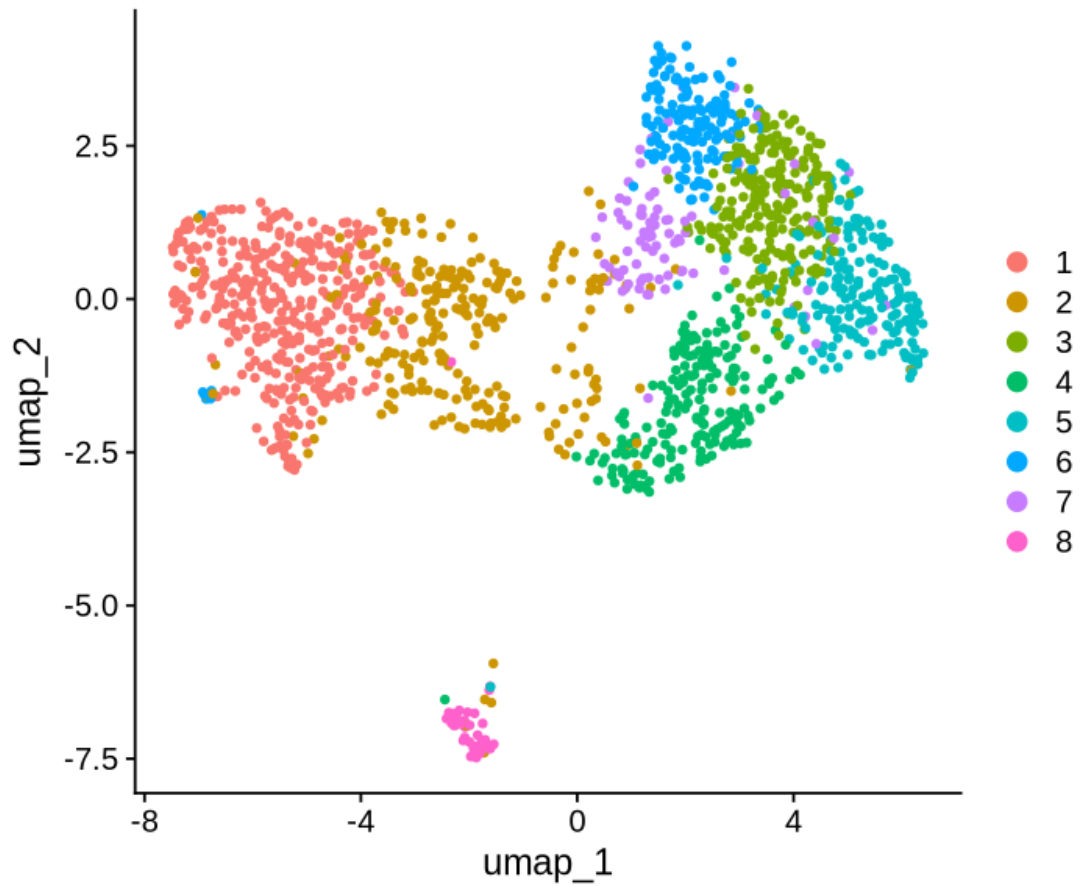
19:38:37 Commencing smooth kNN distance calibration using 1 thread
with target n_neighbors = 30

19:38:45 Initializing from normalized Laplacian + noise (using RSpectra)

19:38:45 Commencing optimization for 500 epochs, with 64758 positive edges

19:38:45 Using rng type: pcg

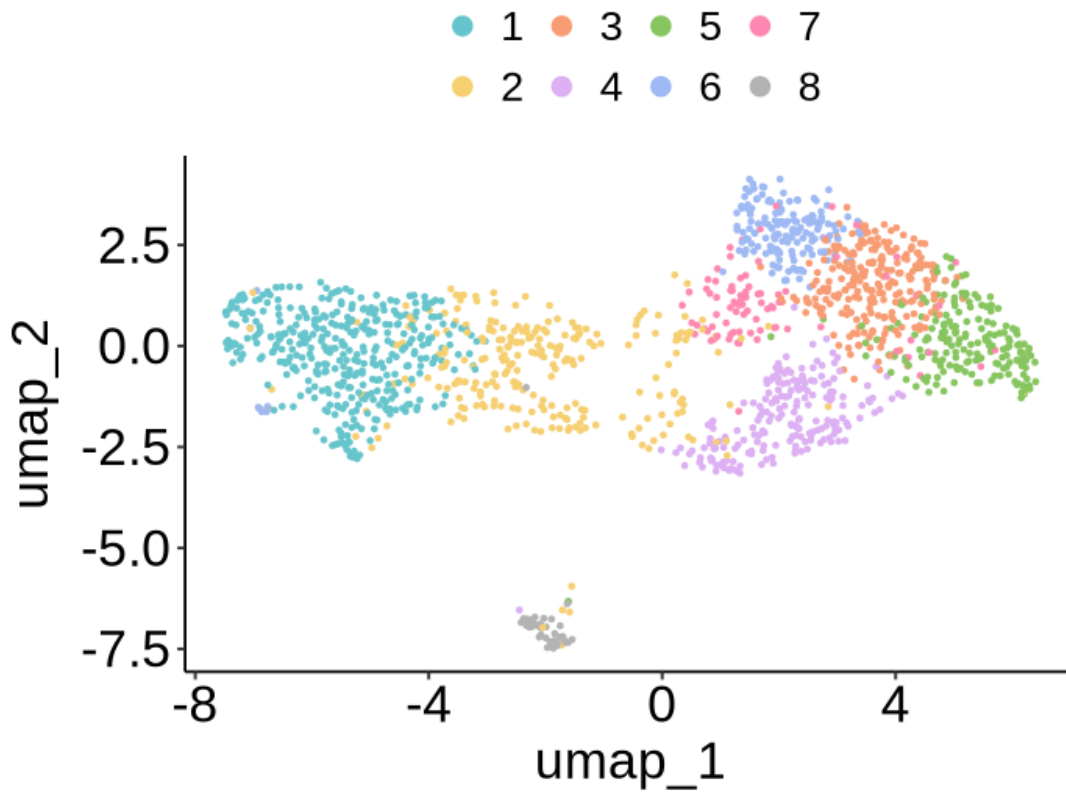
19:38:51 Optimization finished



```
[15]: options(repr.plot.width=6, repr.plot.height=5)

DimPlot(objTE_SoloTE, reduction = "umap", group.by = "seurat_clusters",
        shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  scale_color_carto_d(palette = "Pastel") +
  theme(text=element_text(size=20))
ggsave("figures_10xMouse_RA/umap_locus_clusters.pdf", device = "pdf", width=6, height=5)
```


seurat_clusters

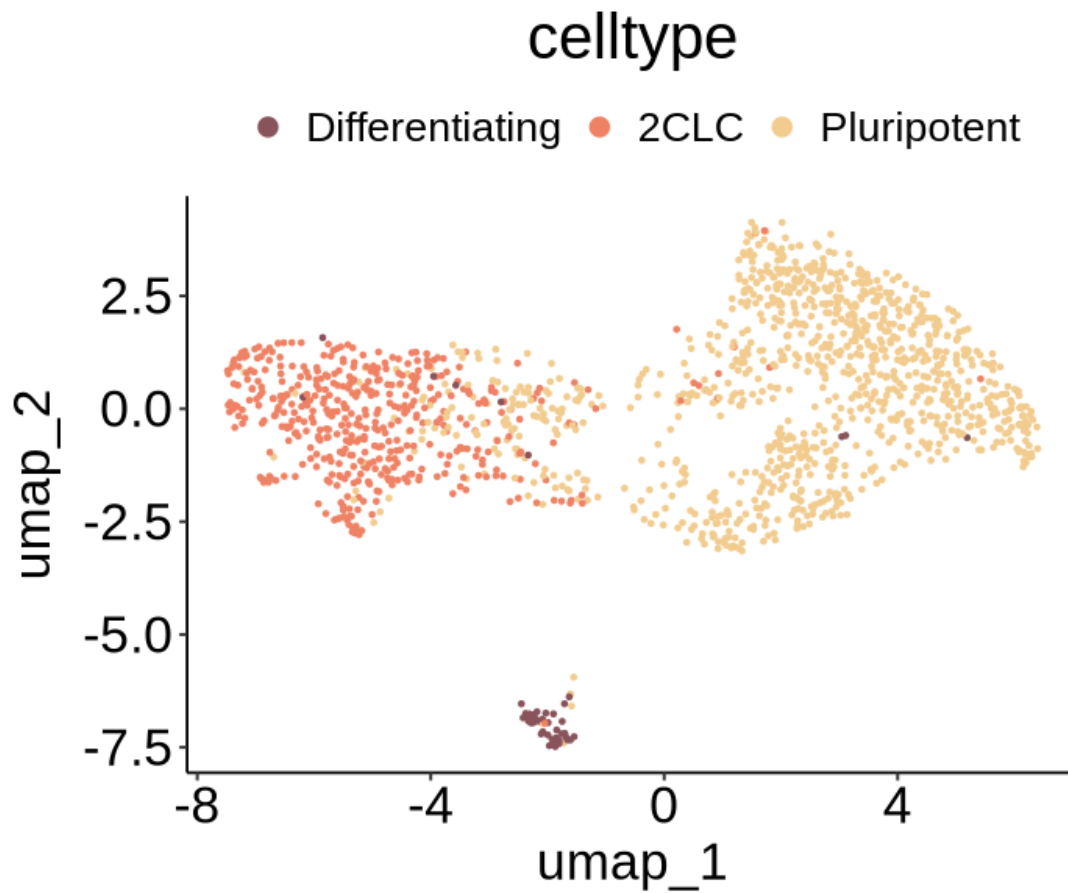


```
[8]: options(repr.plot.width=6, repr.plot.height=5)

# transfer celltype annotation from genes
objTE_SoloTE$celltype <- objGenes_SoloTE$celltype[Cells(objTE_SoloTE)]

options(repr.plot.width=6, repr.plot.height=5)

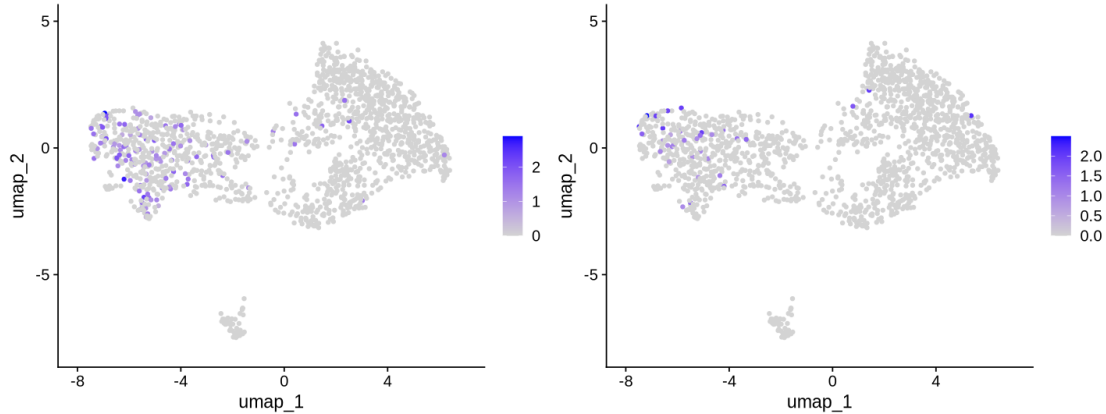
DimPlot(objTE_SoloTE, reduction = "umap", group.by = "celltype",
        cols=colorDict,
        shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  theme(text=element_text(size=20), plot.title=element_text(hjust=0.5))
ggsave(paste0("figures_", dataset_id, "/umap_locus_SoloTE_celltypes.pdf"), device =
  "pdf", width=6, height=5)
```



```
[33]: options(repr.plot.width=12, repr.plot.height=5)

FeaturePlot(objTE_SoloTE, features = c('chr10-105000001-105001301-MERVL-int:
↳ERVL:LTR-1.1--',
    'chr10-107130124-107135454-MERVL-int:ERVL:LTR-1.5-+'))
```

chr10-105000001-105001301-MERV1-int:ERV1:LTR-1.1-chr10-107130124-107135454-MERV1-int:ERV1:LTR-1.5-+



```
[39]: # Comparison to genes

options(repr.plot.width=8, repr.plot.height=8)

objGenes_SoloTE$celltype_num <- objGenes_SoloTE$seurat_clusters
levels(objGenes_SoloTE$celltype_num) <- 1:
  ↪length(levels(objGenes_SoloTE$seurat_clusters))

# Add cluster information from objTE_clustered to objGenes_clustered
objGenes_SoloTE$clusters.0.1 <- objGenes_SoloTE$seurat_clusters
objGenes_SoloTE$clusters.0.2 <- objTE_SoloTE$seurat_clusters
objGenes_SoloTE$clusters.0.3 <- objGenes_SoloTE$celltype

concordanceTable <- table(objGenes_SoloTE$seurat_clusters,
  ↪objTE_SoloTE$seurat_clusters)

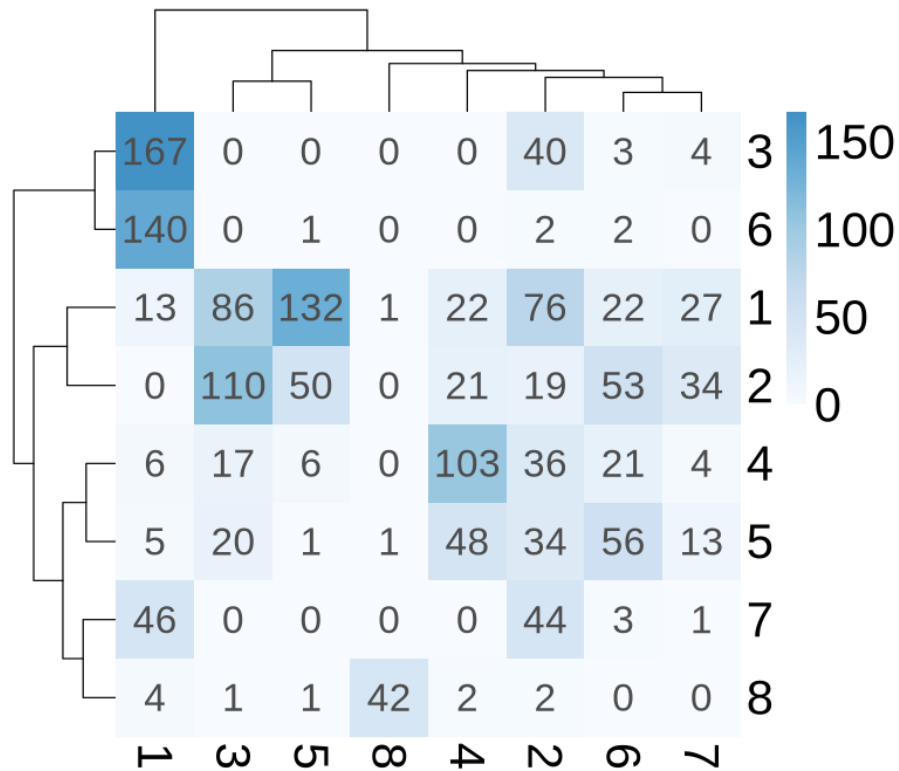
pheatmap(concordanceTable, display_numbers = T, color = colorRampPalette(brewer.
  ↪pal(9, 'Blues')[1:6])(100),
  border_color = NA, number_format = "%.0f", cellwidth = 40, cellheight
  ↪= 40, fontsize = 25)

# Generate the clustree plot
clustree(objGenes_SoloTE, prefix = "clusters.", node_text_angle=0,
  ↪node_text_size=5) + theme(text=element_text(size=15)) +
  scale_color_manual(values=alpha(c("#81B29A", "#8A8BA8", "#F2CC8F"), 0.6)) +
  ↪scale_size(range = c(3,20)) +
  guides(colour = FALSE)
ggsave("figures_10xMouse_RA/clustree_clusters_allcelltypes.pdf")
```

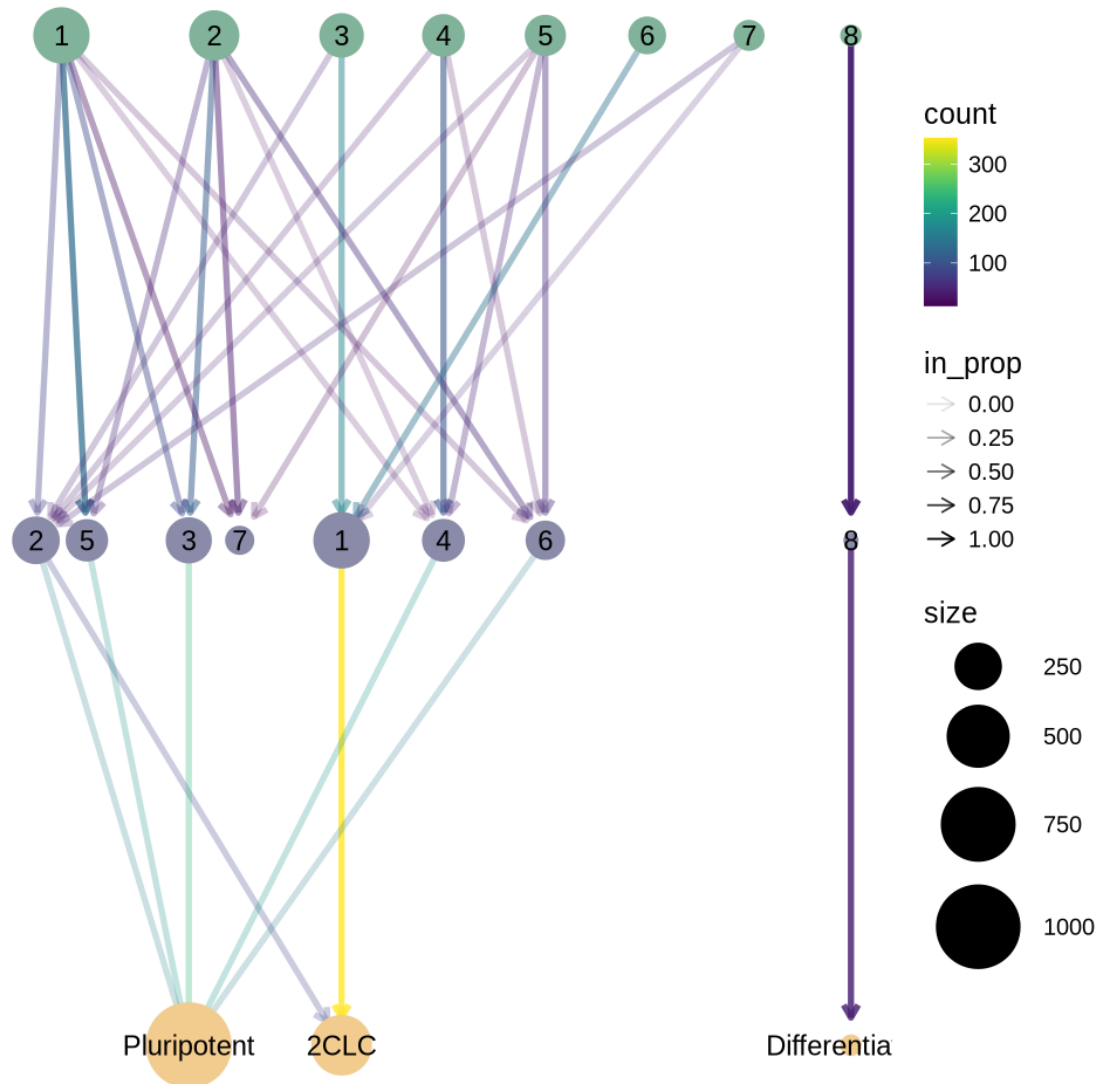
```
objTE_SoloTE$RNAclusters <- objGenes_SoloTE$seurat_clusters
```

Scale for **size** is already present.

Adding another scale for **size**, which will replace the existing scale.



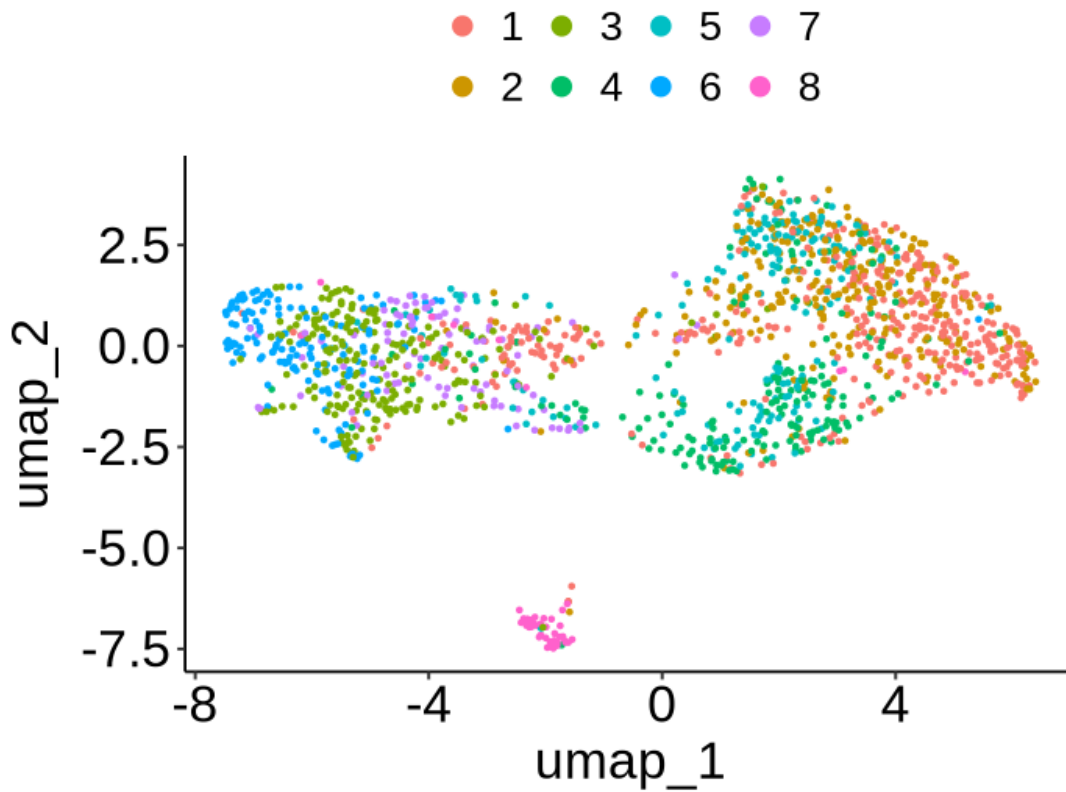
Saving 7 x 7 in image



```
[3]: options(repr.plot.width=6, repr.plot.height=5)

DimPlot(objTE_SoloTE, reduction = "umap", group.by = "RNAclusters",
        shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  theme(text=element_text(size=20))
ggsave("figures_10xMouse_RA/umap_locus_RNAclusters_SoloTE.pdf", device = "pdf")
```

RNAclusters



```
[6]: clustersResList <- list()
      identical(Cells(objGenes_SoloTE), Cells(objTE_SoloTE)) # TRUE, same cells in
      ↪ same order

      for(res in seq(0.5, 2, by=0.1)){
        print(res)

        clusters_genes <- FindClusters(objGenes_SoloTE, resolution = res)
        clusters_TEs <- FindClusters(objTE_SoloTE, resolution = res)

        clustersResList[[as.character(res)]] <-
        ↪ cbind(clusters_genes$seurat_clusters, clusters_TEs$seurat_clusters)
        #
      }
```

TRUE

[1] 0.5

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.8109
Number of communities: 6
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.8073
Number of communities: 6
Elapsed time: 0 seconds
[1] 0.6
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7905
Number of communities: 6
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7857
Number of communities: 6
Elapsed time: 0 seconds
[1] 0.7
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7732
Number of communities: 7
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7684
Number of communities: 8
Elapsed time: 0 seconds
[1] 0.8
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7573
Number of communities: 7
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7520
Number of communities: 8
Elapsed time: 0 seconds
[1] 0.9
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7405
Number of communities: 8
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7356
Number of communities: 9
Elapsed time: 0 seconds
[1] 1
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7253
Number of communities: 8
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7201
Number of communities: 9
Elapsed time: 0 seconds
[1] 1.1
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7115
Number of communities: 10
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7060
Number of communities: 9
Elapsed time: 0 seconds
[1] 1.2
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.6996
Number of communities: 10
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 55968

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6925

Number of communities: 9

Elapsed time: 0 seconds

[1] 1.3

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 49731

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6872

Number of communities: 10

Elapsed time: 0 seconds

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 55968

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6788

Number of communities: 10

Elapsed time: 0 seconds

[1] 1.4

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 49731

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6782

Number of communities: 13

Elapsed time: 0 seconds

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 55968

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6653

Number of communities: 9

Elapsed time: 0 seconds

[1] 1.5

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 49731

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6680

Number of communities: 13

Elapsed time: 0 seconds

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 55968

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6515

Number of communities: 10

Elapsed time: 0 seconds

[1] 1.6

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 49731

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6601

Number of communities: 14

Elapsed time: 0 seconds

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 55968

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6396

Number of communities: 12

Elapsed time: 0 seconds

[1] 1.7

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 49731

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.6513

Number of communities: 13

Elapsed time: 0 seconds

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542

Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.6291
Number of communities: 12
Elapsed time: 0 seconds
[1] 1.8
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.6428
Number of communities: 14
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.6198
Number of communities: 13
Elapsed time: 0 seconds
[1] 1.9
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.6324
Number of communities: 14
Elapsed time: 0 seconds
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.6106
Number of communities: 14
Elapsed time: 0 seconds
[1] 2
Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
 Maximum modularity in 10 random starts: 0.6262
 Number of communities: 16
 Elapsed time: 0 seconds
 Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
 Number of edges: 55968

Running Louvain algorithm...
 Maximum modularity in 10 random starts: 0.6033
 Number of communities: 14
 Elapsed time: 0 seconds

```
[7]: library(mclust)
library(viridis)

resolutions <- seq(0.5, 2.0, by = 0.1)

ari_matrix <- matrix(data=NA, nrow=length(resolutions),
  ↪ncol=length(resolutions))
colnames(ari_matrix) <- paste0("gene_r",resolutions)
rownames(ari_matrix) <- paste0("TE_r",resolutions)

for(r_gene in resolutions){
  for(r_TE in resolutions){
    ari <- adjustedRandIndex(clustersResList[[as.character(r_gene)]][,1],
      clustersResList[[as.character(r_TE)]][,2])
    ari_matrix[paste0("TE_r",r_TE),paste0("gene_r",r_gene)] <- ari
  }
}
ari_matrix

p <- heatmap(ari_matrix, display_numbers = T, color = mako(100, alpha = 1,
  ↪begin = 0, end = 1, direction = 1)[],
  border_color = NA, number_format = "%.3f", number_color="black",
  cluster_rows = FALSE, cluster_columns = FALSE,
  cellwidth = 25, cellheight = 25)

pdf(paste0("figures_",dataset_id,"/ARI_celltypes_SoloTE.pdf"), width=9.5,
  ↪height=9)
p
dev.off()
# ari_values <- sapply(resolutions, function(r) {
#   adjustedRandIndex(clustersResList[[as.character(r)]][,1],
#   clustersResList[[as.character(r)]][,2])
```

```
# })

# plot(resolutions, ari_values, type = "b",
#       xlab = "Resolution", ylab = "ARI",
#       main = "Gene vs Transposon Clustering Similarity")
```

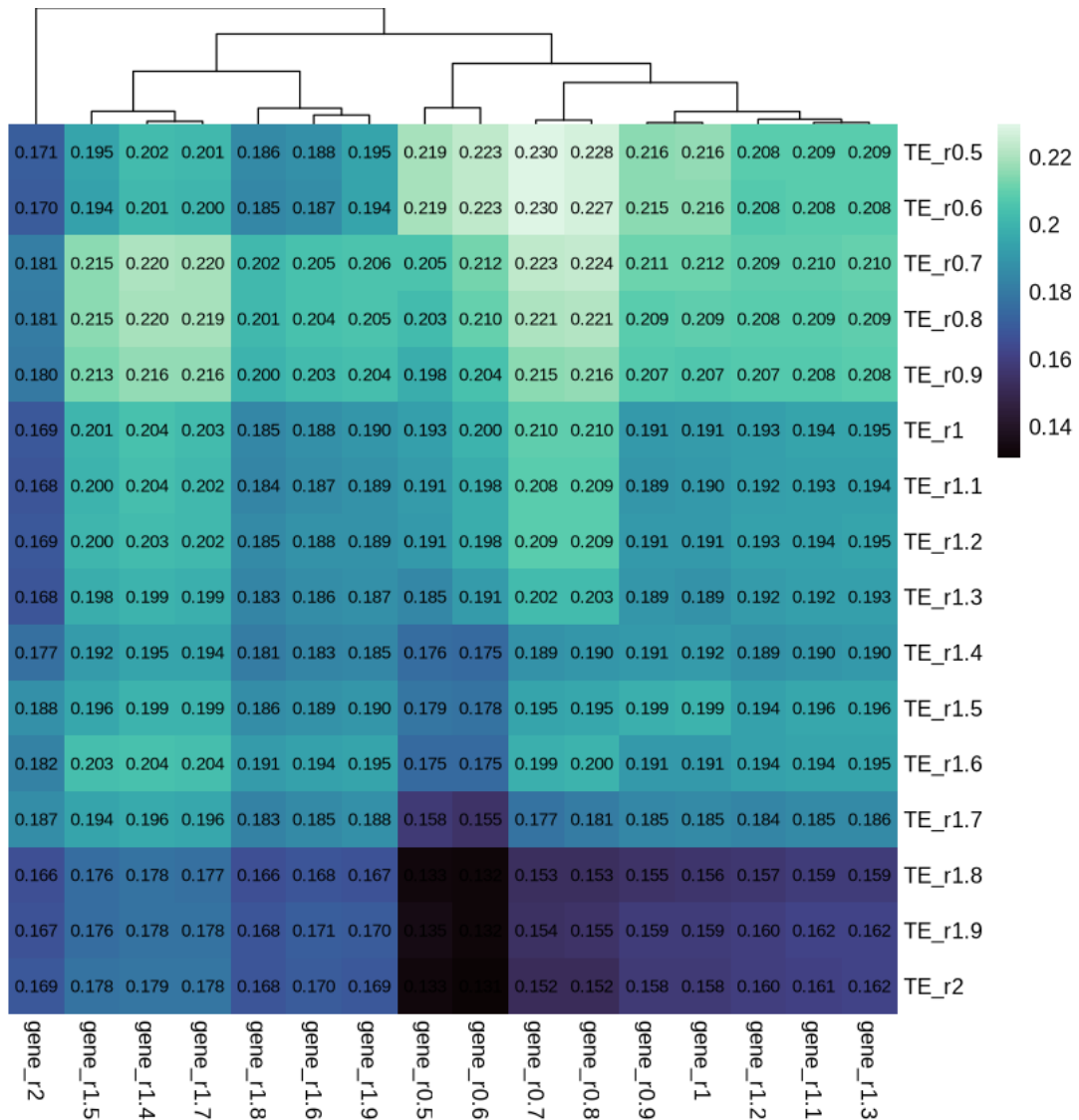
Package 'mclust' version 6.1.1

Type 'citation("mclust")' for citing this R package in publications.

Loading required package: viridisLite

	gene_r0.5	gene_r0.6	gene_r0.7	gene_r0.8	gene_r0.9	gene_r1
TE_r0.5	0.2191505	0.2229374	0.2297387	0.2276990	0.2157145	0.2160181
TE_r0.6	0.2193259	0.2229464	0.2296239	0.2274096	0.2152267	0.2155391
TE_r0.7	0.2052594	0.2123646	0.2233473	0.2241175	0.2111831	0.2115381
TE_r0.8	0.2027788	0.2100201	0.2207828	0.2214720	0.2088189	0.2091761
TE_r0.9	0.1976314	0.2044404	0.2152398	0.2159340	0.2072798	0.2069111
TE_r1	0.1925005	0.1999808	0.2095184	0.2097706	0.1905408	0.1914841
TE_r1.1	0.1913819	0.1979626	0.2084675	0.2087271	0.1894764	0.1904211
TE_r1.2	0.1910436	0.1976438	0.2085964	0.2087746	0.1906008	0.1909321
TE_r1.3	0.1850874	0.1911787	0.2023034	0.2025153	0.1891707	0.1888481
TE_r1.4	0.1759103	0.1746916	0.1887994	0.1899228	0.1908577	0.1915561
TE_r1.5	0.1785930	0.1777364	0.1948697	0.1953868	0.1989150	0.1992811
TE_r1.6	0.1748722	0.1746716	0.1985023	0.1999487	0.1910116	0.1909391
TE_r1.7	0.1581889	0.1552954	0.1774657	0.1805331	0.1851290	0.1854661
TE_r1.8	0.1334770	0.1320365	0.1530285	0.1532441	0.1552931	0.1557561
TE_r1.9	0.1348142	0.1320087	0.1538000	0.1546259	0.1585287	0.1588501
TE_r2	0.1330255	0.1307774	0.1517366	0.1523820	0.1581174	0.1580161

pdf: 2



```
[61]: res <- "0.8"
TEclusters <- clustersResList[[res]][,2]
table(clustersResList[[res]][,2])
objTE_SoloTE <- FindClusters(objTE_SoloTE, resolution = as.numeric(res))

res <- "1"
GENEclusters <- clustersResList[[res]][,1]
table(clustersResList[[res]][,1])
objGenes_SoloTE <- FindClusters(objGenes_SoloTE, resolution = as.numeric(res))

objTE_SoloTE$RNAclusters <- objGenes_SoloTE$seurat_clusters
objGenes_SoloTE$TEclusters <- objTE_SoloTE$seurat_clusters
```

```

concordanceTable <- table(objGenes_SoloTE$seurat_clusters,
  ↪objTE_SoloTE$seurat_clusters)
rownames(concordanceTable) <- paste("GeneCluster_", rownames(concordanceTable))
colnames(concordanceTable) <- paste("TECluster_", colnames(concordanceTable))

```

```

  1   2   3   4   5   6   7   8
388 315 256 192 155 117  77  42

```

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 55968

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7520
Number of communities: 8
Elapsed time: 0 seconds

```

  1   2   3   4   5   6   7   8
382 267 216 203 173 155  94  52

```

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1542
Number of edges: 49731

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.7253
Number of communities: 8
Elapsed time: 0 seconds

```

[62]: options(repr.plot.width=6, repr.plot.height=5)

DimPlot(objTE_SoloTE, reduction = "umap", group.by = "seurat_clusters",
  shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  scale_color_carto_d(palette = "Pastel") +
  theme(text=element_text(size=20))
ggsave("figures_10xMouse_RA/umap_locusTE_SoloTE_seurat_clusters.pdf", device =
  ↪"pdf")

DimPlot(objTE_SoloTE, reduction = "umap", group.by = "RNAclusters",
  shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  scale_color_brewer(palette="Spectral") +

```



```

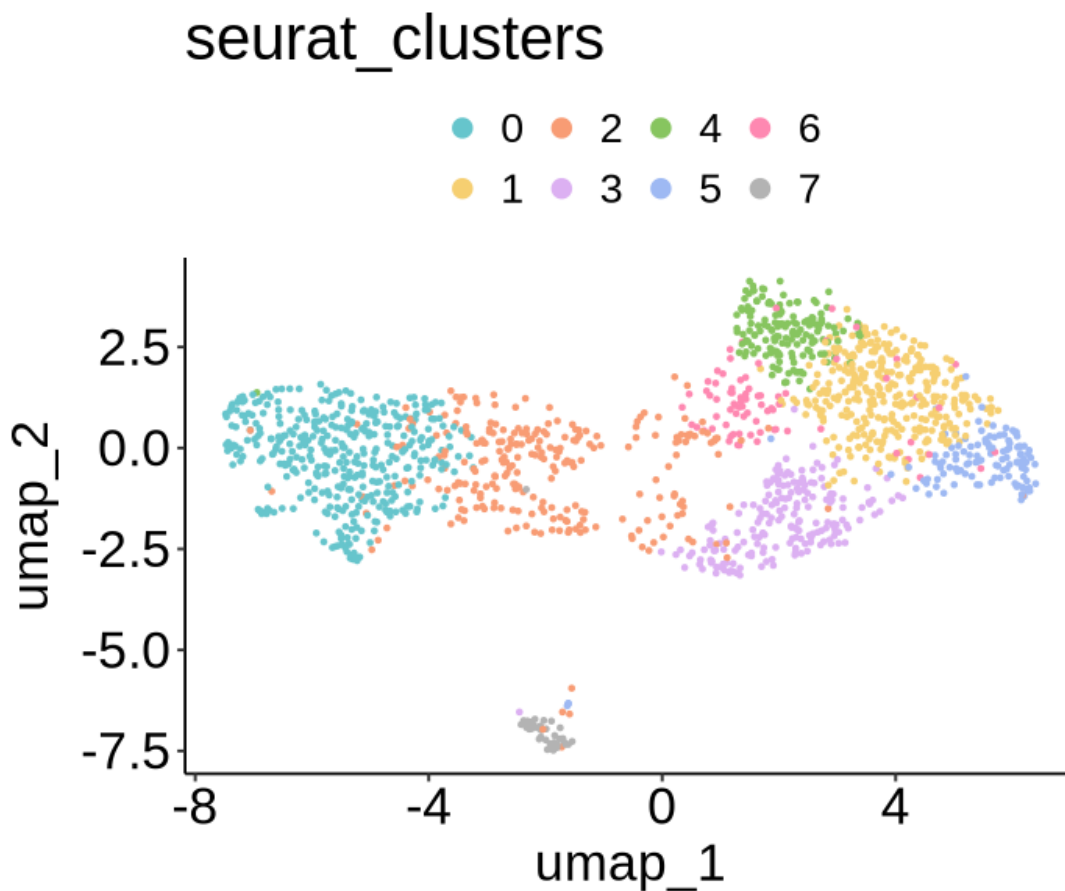
    theme(text=element_text(size=20))
ggsave("figures_10xMouse_RA/umap_locustTE_SoloTE_RNAclusters.pdf", device = "pdf")

DimPlot(objGenes_SoloTE, reduction = "umap", group.by = "TEclusters",
        shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  scale_color_carto_d(palette = "Pastel") +
  theme(text=element_text(size=20))
ggsave("figures_10xMouse_RA/umap_genes_SoloTE_TEclusters.pdf", device = "pdf")

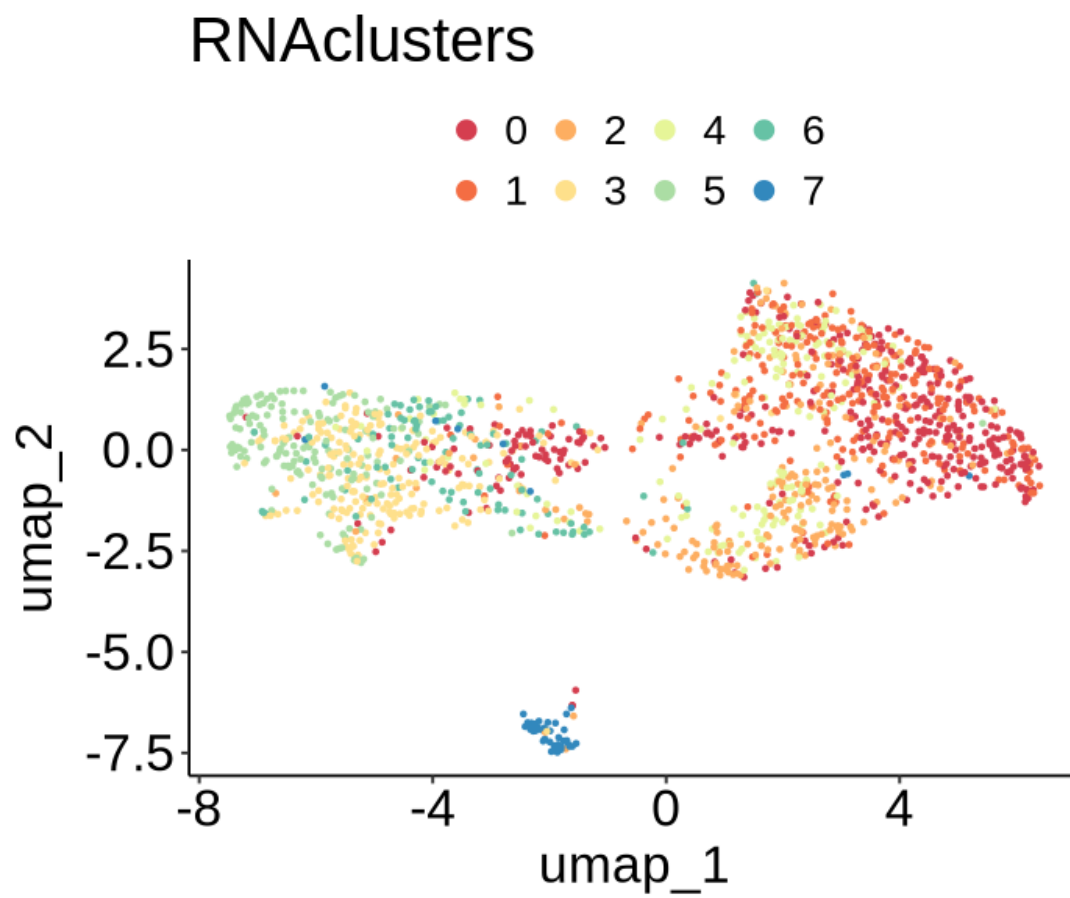
DimPlot(objGenes_SoloTE, reduction = "umap", group.by = "seurat_clusters",
        shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  scale_color_brewer(palette="Spectral") +
  theme(text=element_text(size=20))
ggsave("figures_10xMouse_RA/umap_genes_SoloTE_RNAclusters.pdf", device = "pdf")

```

Saving 7 x 7 in image

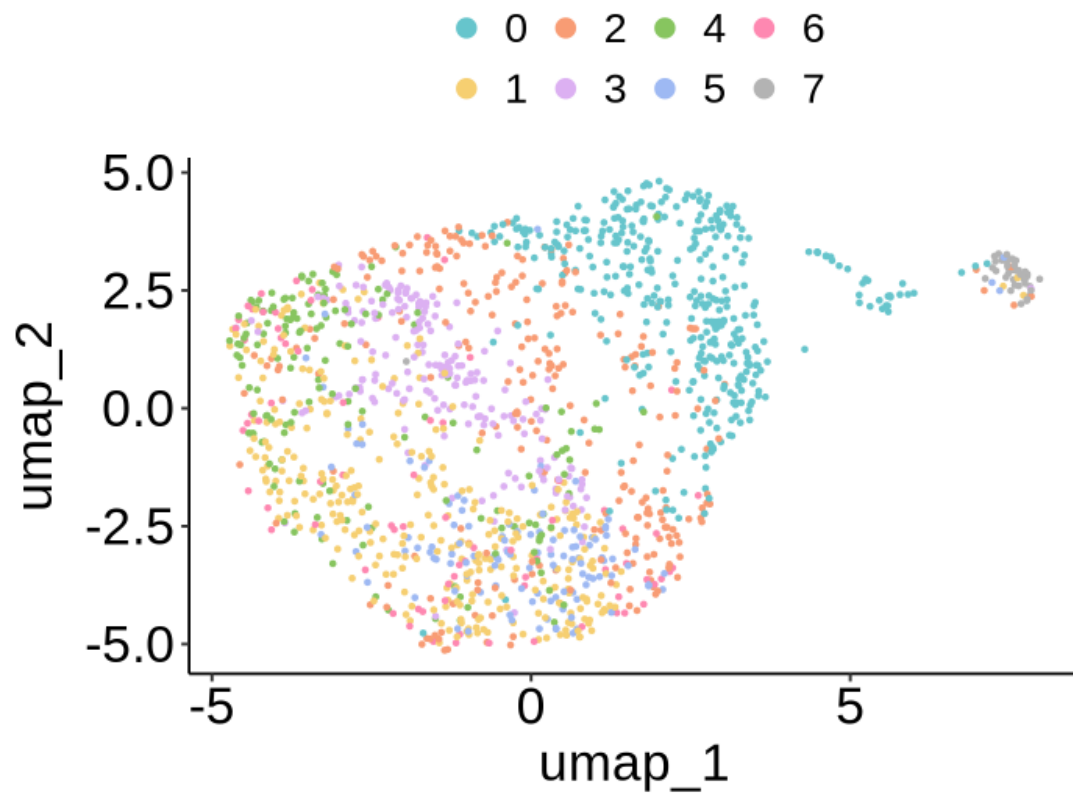


Saving 7 x 7 in image



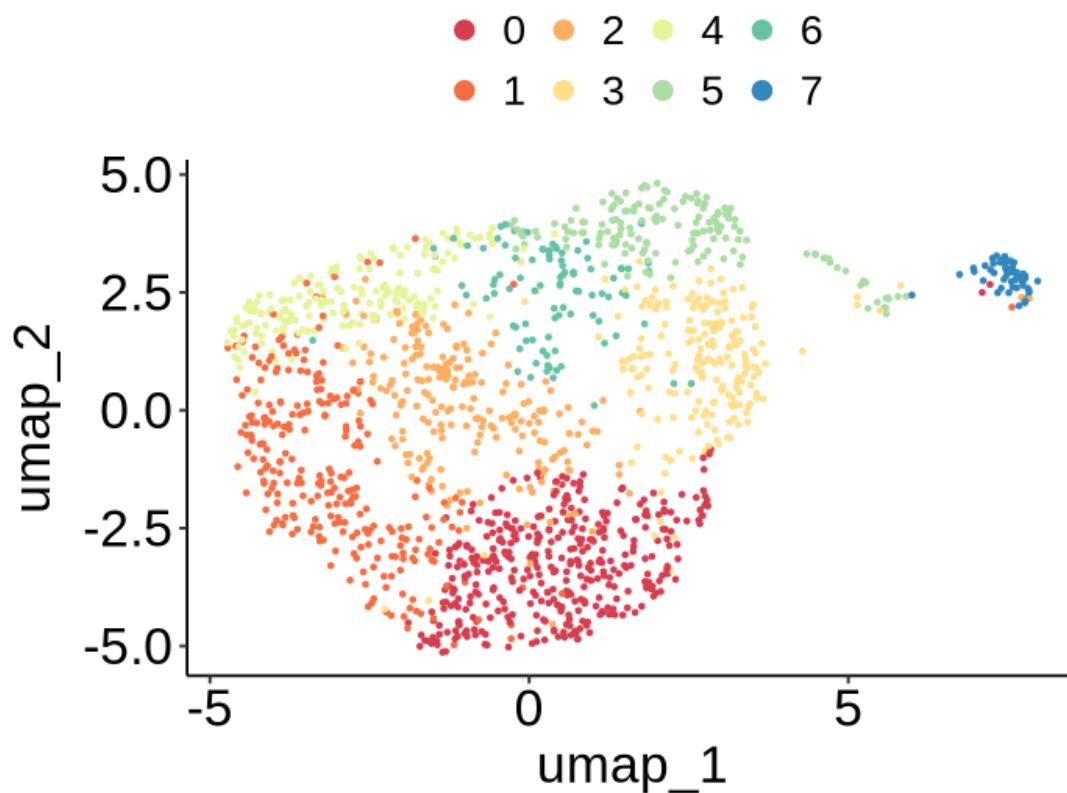
Saving 7 x 7 in image

TEclusters



Saving 7 x 7 in image

seurat_clusters



```
[64]: library(clue)
library(grid)
options(repr.plot.width=9, repr.plot.height=8)

# Suppose your matrix has more rows than columns
nr <- nrow(concordanceTable)
nc <- ncol(concordanceTable)

if(nr > nc){
  # pad with zeros to make it square
  M <- cbind(concordanceTable, matrix(0, nrow = nr, ncol = nr - nc))
} else {
  M <- rbind(concordanceTable, matrix(0, nrow = nc - nr, ncol = nc))
}

perm <- solve_LSAP(M, maximum = TRUE)

# keep only the original columns
```

```

M_reordered <- M[, perm[1:nc]]
rownames(M_reordered)

ph <- pheatmap( M_reordered[(rownames(M_reordered)!=""),!is.
  ↪na(colnames(M_reordered))], ,
  display_numbers = T, color = colorRampPalette(brewer.pal(9,'Blues'))[1:
  ↪7])(100),#adjustcolor((colorRampPalette(carto_pal(name="Emrld")))(100),alpha.
  ↪f=1),
  main = "Cluster concordance - TEs VS Genes",
  cluster_rows=FALSE, cluster_cols=FALSE,
  fontsize_number = 14, fontsize = 14, annotation_names_row =
  ↪TRUE, number_color = "black",
  border_color = NA, number_format = "%.0f", cellwidth = 30, cellheight
  ↪= 30)

# write to PDF safely

pdf(paste0("figures_", dataset_id, "/"
  ↪heatmap_cluster_clusters_concordance_SoloTE.pdf"), width=8, height=9)

grid::grid.newpage()
grid::grid.draw(ph$gtable)

dev.off()

```

1. 'GeneCluster__0' 2. 'GeneCluster__1' 3. 'GeneCluster__2' 4. 'GeneCluster__3' 5. 'GeneCluster__4' 6. 'GeneCluster__5' 7. 'GeneCluster__6' 8. 'GeneCluster__7'

pdf: 2

Cluster concordance - TEs VS Genes

81	137	23	16	23	25	77	0	GeneCluster_0	150
26	122	14	0	53	30	22	0	GeneCluster_1	100
7	36	105	5	22	7	33	1	GeneCluster_2	50
0	0	0	160	1	3	39	0	GeneCluster_3	0
0	18	49	8	53	12	33	0	GeneCluster_4	
1	0	0	152	1	0	1	0	GeneCluster_5	
0	0	0	43	2	0	49	0	GeneCluster_6	
2	2	1	4	0	0	2	41	GeneCluster_7	
TECluster_5	TECluster_1	TECluster_3	TECluster_0	TECluster_4	TECluster_6	TECluster_2	TECluster_7		

```
[41]: ## Save objects
# saveRDS(objGenes_SoloTE, paste0("data_", dataset_id, "/soloTE_", dataset_id,
# ↪ "_GENE_seuratObj_251124.RDS"))
# saveRDS(objTE_SoloTE, paste0("data_", dataset_id, "/soloTE_", dataset_id,
# ↪ "_seuratObj_251124.RDS"))
```

4 Checkpoint

```
[13]: #save.image("workspaces/10xMouse_RA_SoloTE_251124.Rdata")
load("workspaces/10xMouse_RA_SoloTE_251124.Rdata")
```

```
[12]: options(repr.plot.width=8, repr.plot.height=4)
library(reshape2)
concordanceTable <- table(objTE_SoloTE$seurat_clusters, objTE_SoloTE$celltype)
```

```

df <- melt(concordanceTable)
df <- df[df$value!=0,]
colnames(df) <- c("cluster", "celltype", "nCells")
df$cluster <- as.character(df$cluster)
df$clusterSize <- table(objTE_SoloTE$seurat_clusters)[df$cluster]
df$percentage <- as.numeric(df$nCells / df$clusterSize *100)

# df <- df %>%
#   group_by(cluster) %>%
#   mutate(main = celltype[which.max(percentage)]) %>%
#   ungroup() %>%
#   arrange(main, desc(percentage))
# df$cluster <- factor(df$cluster, levels = unique(df$cluster))

ggplot(df, aes(x=cluster, y=nCells, fill=celltype)) +
  geom_col() +
  xlab("TE cluster") +
  scale_fill_manual(values=colorDict)+
  theme_minimal() + theme(text=element_text(size=20))
ggsave(paste0("figures_", dataset_id, "/"
  ↪clusterIdentity_barplot_SoloTE_noplatelet.pdf"), width=8.5, height=4)

```

